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# Systemic Therapy in Patients With Metastatic Castration-Resistant Prostate Cancer: ASCO Guideline Update

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## ABSTRACT

*ASCO Guidelines provide recommendations with comprehensive review and analyses of the relevant literature for each recommendation, following the guideline development process as outlined in the ASCO Guidelines Methodology Manual. ASCO Guidelines follow the ASCO Conflict of Interest Policy for Clinical Practice Guidelines.*

*Clinical Practice Guidelines and other guidance ("Guidance") provided by ASCO is not a comprehensive or definitive guide to treatment options. It is intended for voluntary use by clinicians and should be used in conjunction with independent professional judgment. Guidance may not be applicable to all patients, interventions, diseases or stages of diseases. Guidance is based on review and analysis of relevant literature, and is not intended as a statement of the standard of care. ASCO does not endorse third-party drugs, devices, services, or therapies and assumes no responsibility for any harm arising from or related to the use of this information. See complete disclaimer in Appendix 1 and 2 (online only) for more.*

**PURPOSE** To provide evidence-based recommendations for patients with metastatic castration-resistant prostate cancer (mCRPC).

**METHODS** An Expert Panel including patient representation completed a systematic review of the evidence and made recommendations.

**RESULTS** Depending upon prior treatment received, androgen receptor pathway inhibitors (ARPIs: enzalutamide, abiraterone with prednisone), poly(ADP-ribose) polymerase inhibitors (PARPi), chemotherapeutic agents (docetaxel, cabazitaxel), radiopharmaceuticals (radium 223, $^{177}$Lu-prostate-specific membrane antigen [PSMA]-617), and sipuleucel-T have demonstrated an overall survival (OS) benefit for patients with mCRPC. For patients with *BRCA1*/2 alterations who did not receive prior ARPI, the combination of PARPi and ARPI (talazoparib + enzalutamide, olaparib and/or niraparib + abiraterone) has shown clinical benefit. For patients with *BRCA1*/2 alterations who received prior ARPI or ARPI followed by docetaxel, olaparib showed OS benefit. In select patients with microsatellite instability-high/mismatch repair-deficient, pembrolizumab showed clinical efficacy.

**RECOMMENDATIONS** Prior systemic therapy for castration-sensitive prostate cancer will determine subsequent therapy used for mCRPC. Continue androgen-deprivation therapy for patients with mCRPC indefinitely. Early adoption of somatic genetic testing and palliative care is recommended. Patients with mCRPC and bony metastases should receive a bone-protective agent. The panel recommends the combination of ARPI with PARPi in patients with *BRCA1*/2 alterations who did not receive prior ARPI. For patients who received prior ARPI, the panel recommends docetaxel chemotherapy. The panel recommends $^{177}$Lu-PSMA-617 or cabazitaxel chemotherapy for patients who receive prior ARPI and docetaxel chemotherapy. For patients with *BRCA1*/2 alterations who received prior ARPI, the panel recommends PARPi monotherapy. Radium 223 is recommended for patients with symptomatic bone–only disease. Evidence for optimal sequencing for mCRPC regimens is lacking.

Additional information is available at www.asco.org/genitourinary-cancer-guidelines.

ACCOMPANYING CONTENT

Listen to the podcast by Dr Garje and Brittany Harvey at https://ascopubs.org/do/systemic-therapy-patients-metastatic-castration-resistant-prostate-cancer-guideline

Appendix

Data Supplement

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## TARGET POPULATION AND AUDIENCE

**Target Population**
Patients with metastatic castration-resistant prostate cancer (mCRPC).

**Target Audience**
Patients with mCRPC, oncologists, primary care physicians, nuclear medicine clinicians, nurses, social workers, counsellors, and other care givers.

## INTRODUCTION

Metastatic prostate cancer is a major cause of disease burden in patients, representing an estimated 299,010 new cases (the highest incidence in males, representing 29% of all new cases) and being responsible for 35,250 deaths (the second highest cause of mortality in males, representing 11% of all cancer-related deaths) in 2024.[1]

A earlier version of this guideline was published in 2014,[2] and since that time, two focused updates were completed, one in 2022,[3] and one in 2023.[4] The purpose of this guideline update is to gather and evaluate the best available evidence on systemic treatment options for patients with metastatic castration-resistant prostate cancer (mCRPC) and to provide a series of recommendations to inform their treatment. This guideline will address questions on diagnostics, genomic testing, treatment, and specifically treatment options for small cell, neuroendocrine, and poorly differentiated prostate cancer.

## GUIDELINE QUESTIONS

This clinical practice guideline addresses three overarching clinical questions:

1. What are the treatment options for patients for mCRPC? Role of systemic therapy should be stratified based on prior therapy received in castration-sensitive and M0 CRPC:
   1.1. Prior androgen-deprivation therapy (ADT) alone
   1.2. Prior ADT + androgen receptor pathway inhibitor (ARPI)
   1.3. Prior ADT + docetaxel
   1.4. Prior ADT + docetaxel + ARPI
2. What are the treatment options for de novo or treatment-emergent small cell neuroendocrine carcinoma of the prostate?
   2.1 Cisplatin or carboplatin + etoposide
   2.2 Carboplatin + cabazitaxel
   2.3 Clinical trials
3. How to assess for response while on systemic therapy for mCRPC?
   3.1 Lab assessment with prostate-specific antigen (PSA) levels, testosterone to ensure castrate levels, and alkaline phosphatase (AP)
   3.2 What scans (computed tomography [CT] chest, abdomen, and pelvis [CAP], bone scan, molecular—$^{68}$Ga, $^{18}$F piflufolastat [PYL], $^{18}$F flotufolastat prostate-specific membrane antigen [PSMA], and choline and fluciclovine imaging) are recommended for response assessment? At what frequency should these scans occur?

## METHODS

The systematic review informing the guideline product was conducted using the living interactive evidence (LIvE) synthesis framework[66] and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

The guideline product was developed by a multidisciplinary expert panel, which included a patient representative and an ASCO guidelines staff member with health research methodology expertise (Appendix Table A1).

### Literature Search

The search strategy was designed by an experienced medical librarian with input from the principal investigator. A comprehensive search of Ovid MEDLINE and Epub Ahead of Print, In-Process & Other Non-Indexed Citations, and Daily; Ovid EMBASE; Ovid Cochrane Central Register of Controlled Trials; and Ovid Cochrane Database of Systematic Reviews was conducted initially from each database's inception through June 16, 2021.

Subsequently, a living search has been created with weekly updates to identify new evidence as it becomes available.

### Guideline Development Process

#### *Study Selection*

Full-text articles of phase II or III randomized clinical trials (RCTs) were included based on the following criteria:

- Population: patients with mCRPC
- Interventions: systemic treatment options
- Comparator: any comparable systemic treatment option
- Outcomes: overall survival (OS), radiographic and/or PSA progression-free survival (PFS) or disease-free survival, quality of life, adverse effects

Articles were excluded from the systematic review if they were (1) meeting abstracts not subsequently published in peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, and narrative reviews; and (3) published in a non-English language. A manual search was also conducted to identify articles that were not included as part of the systematic review search but were deemed eligible for inclusion to inform the current guideline.

Ten full-panel meetings were held, and members were asked to provide ongoing input on the guideline development protocol, quality and assessment of the evidence, generation of recommendations, draft content, as well as review and approve drafts during the entire development of the guideline. ASCO staff met routinely with the Expert Panel co-chairs and corresponded with the Panel via e-mail to coordinate the process to completion. Ratings for the strength of the recommendation and evidence quality are provided with each recommendation, defined in Appendix Table A2. The quality of the evidence for each outcome was assessed using the Cochrane Risk of Bias tool and elements of the grading of recommendation, assessment, development, and evaluation (GRADE) quality assessment and recommendations development process.[5,6] GRADE quality assessment labels (ie, high, moderate, low, very low) were assigned for each outcome by the project methodologist in collaboration with the Expert Panel co-chairs and reviewed by the full Expert Panel. All funding for the administration of the project was provided by ASCO.

### *Data Extraction and Quality Assessment*

The extracted data included but was not limited to trial characteristics, baseline population characteristics, and outcome results in the overall population and in clinically relevant subgroups. Two reviewers (S.A.A.N. and I.B.R.) independently extracted data with machine-facilitated annotations from the included trials and examined risk of bias in each trial qualitatively using the Cochrane Risk of Bias tool version 2. Discrepancies in the process were resolved by consensus and input from a third reviewer.

### *Meta-Analysis*

A random-effects DerSimonian-Laird meta-analysis was only conducted to pool effect estimates from >1 trials where the population, intervention, comparator, and outcomes were considered homogeneous. In instances where within trial pooling was needed to combine the treatment effects of homogeneous subgroups, a fixed-effect model was used. Survival outcomes were expressed as hazard ratio (HR) with 95% CIs and raw binary outcomes were expressed as relative risk and 95% CI.

## Guideline Review and Approval

The draft recommendations were released to the public for open comment from August 23, 2024, through September 6, 2024. Response categories of "Agree as written," "Agree with suggested modifications," and "Disagree. See comments" were captured for every proposed recommendation with 59 written comments received. A total of 66% of the 264 responses either agreed or agreed with slight modifications to the recommendations and 5% of the respondents disagreed. Expert Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions.

A guideline implementability review was conducted. Based on this review, revisions were made to the draft to clarify recommended actions for clinical practice.

All changes were incorporated into the final manuscript prior to ASCO Evidence Based Medicine Committee (EBMC) review and approval. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO EBMC before submission to the *Journal of Clinical Oncology* for editorial review and consideration for publication.

## Guideline Updating

The ASCO Expert Panel and guidelines staff will work with co-chairs to keep abreast of any substantive updates to the guideline. Based on formal review of the emerging literature, ASCO will determine the need to update. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the guideline update process. This is the most recent information as of the publication date.

# RESULTS

As of September 1, 2024, 33,365 references have been retrieved. Of these, a total 141 unique trials with 181 references have been included in the systematic review[66] as shown in the Data Supplement (Fig S1: PRISMA flowchart outlining the study selection process, online only). Among the pool of eligible trials included in the systematic review, 22 trials[67-126] informed the current guideline. Of these 22 trials, two trials ARTO[100] and KEYNOTE 158[95,96] were manually identified and included to inform this guideline as per the Expert Panel consensus.

## Characteristics of Studies Identified in the Literature Search

A summary of characteristics of the included trials is outlined in Table 1. A total of 14,707 patients were included in the 22 trials[67-126] selected in this guideline. Median age for these patients was 70 (IQR, 69-71) years. Nineteen trials were phase III and three trials were phase II. Radiographic PFS (rPFS) was reported as the primary end point in 10 trials. Similarly, OS was also the primary outcome in 10 trials. Time to tumor progression was reported in two trials, and composite PFS, radiographic objective response rate, and PSA50 response were each reported in one trial. Sixteen trials reported data on health-related quality of life. Detailed population characteristics along with the results for health-related quality of life assessments at the level of each study are outlined in the Data Supplement (Table S1: Population characteristics and Table S2: Summary of quality of life outcomes for included trials). Distribution of race and ethnicity across the included trials is shown in the Data Supplement (Table S3: Distribution of race and ethnicity across the included trials). In terms of the quality of the included trials, the risk of bias was low for OS across all domains. However, five trials had some concerns for

**TABLE 1.** Summary of Characteristics of Included Trials

| Trial | Year | Treatment Arm | Control Arm | Phase | Arms | Estimated Accrual, No. | Actual Accrual, No. | Participants, No. | Age, Years, Median (range) | Primary End Point | QoL Reported | Guideline Recommendation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MAGNITUDE[67-72] | 2023 | Niraparib + abiraterone | Abiraterone | III | 2 | 400 | 423 | Rx: 212<br>Ctrl: 211 | 69 (43-100) | rPFS | Yes | 1.1.1<br>1.3.1 |
| TALAPRO-2[73-77] | 2023 | Talazoparib + enzalutamide | Enzalutamide | III | 2 | 750 | 805 | Rx: 402<br>Ctrl: 403 | Rx: 71 (66-76)[a]<br>Ctrl: 71 (65-76)[a] | rPFS | Yes | 1.1.1<br>1.1.2<br>1.3.1 |
| PROpel[78-84] | 2023 | Olaparib + abiraterone | Abiraterone | III | 2 | 720 | 796 | Rx: 399<br>Ctrl: 397 | Rx: 69 (43-91)<br>Ctrl: 70 (46-88) | rPFS | Yes | 1.1.1<br>1.3.1 |
| COU-AA-302[85-87] | 2013 | Abiraterone | Placebo | III | 2 | 1,000 | 1,088 | Rx: 546<br>Ctrl: 542 | Rx: 71 (44-95)<br>Ctrl: 70 (44-90) | rPFS OS | Yes | 1.1.3 |
| PREVAIL[88-90] | 2014 | Enzalutamide | Placebo | III | 2 | 1,680 | 1,717 | Rx: 872<br>Ctrl: 845 | Rx: 72 (43-93)<br>Ctrl: 71 (42-93) | rPFS OS | Yes | 1.1.3 |
| TAX 327[91,92] | 2004 | Docetaxel[b] | Mitoxantrone | III | 3 | 1,002 | 1,006 | Arm1: 335<br>Arm2: 334<br>Ctrl: 337 | Arm1: 68 (42-92)<br>Arm2: 69 (36-92)<br>Ctrl: 68 (43-86) | OS | Yes | 1.1.3<br>1.2.2.1 |
| ALSYMPCA[93,94] | 2013 | Radium 223 | Placebo | III | 2 | 900 | 921 | Rx: 614<br>Ctrl: 307 | Rx: 71 (49-90)<br>Ctrl: 71 (44-94) | OS | Yes | 1.1.4<br>1.2.2.1<br>1.3.3<br>1.4.3 |
| KEYNOTE-158[95,96] | 2019 | Pembrolizumab | NA | II | 1 | NA | 233 | 6 (prostate cancer) | 60 (20-87) | ORR | Yes | 1.1.5<br>1.3.3<br>1.4.3 |
| D9901[97,98] | 2006 | Sipuleucel-T | Placebo | III | 2 | 120 | 127 | Rx: 82<br>Ctrl: 45 | Rx: 73 (47-85)<br>Ctrl: 71 (50-86) | TTP | No | 1.1.6<br>1.2.3 |
| IMPACT[99] | 2010 | Sipuleucel-T | Placebo | III | 2 | 500 | 512 | Rx: 341<br>Ctrl: 171 | Rx: 72 (49-91)<br>Ctrl: 70 (40-89) | OS | No | 1.1.6<br>1.2.3 |
| D9902A[98] | 2009 | Sipuleucel-T | Placebo | III | 2 | NA | 98 | Rx: 65<br>Ctrl: 33 | Rx: 70 (51-84)<br>Ctrl: 71 (57-87) | TTP | No | 1.1.6<br>1.2.3 |
| ARTO[100] | 2023 | Abiraterone + SBRT | Abiraterone | II | 2 | 174 | 157 | Rx: 75<br>Ctrl: 82 | Rx: 74 (68-79)[a]<br>Ctrl: 74 (68-79)[a] | PSA50 response | No | 1.1.7<br>1.2.3 |
| PROfound[101-106] | 2020 | Olaparib | Enzalutamide + abiraterone | III | 2 | 240 | 387 | Rx: 256<br>Ctrl: 131 | Rx: 69 (47-91)<br>Ctrl: 69 (49-87) | rPFS | No | 1.2.1<br>1.2.2<br>1.4.2 |
| COU-AA-301[107-112] | 2011 | Abiraterone | Placebo | III | 2 | 1,158 | 1,195 | Rx: 797<br>Ctrl: 398 | Rx: 69 (42-95)<br>Ctrl: 69 (39-90) | OS | Yes | 1.3.2 |
| AFFIRM[113-115] | 2012 | Enzalutamide | Placebo | III | 2 | 1,170 | 1,199 | Rx: 800<br>Ctrl: 399 | Rx: 69 (41-92)<br>Ctrl: 69 (49-89) | OS | Yes | 1.3.2 |
| TROPIC[116,117] | 2010 | Cabazitaxel | Mitoxantrone | III | 2 | 720 | 755 | Rx: 378<br>Ctrl: 377 | Rx: 67 (61-72)[a]<br>Ctrl: 68 (62-73)[a] | OS | Yes | 1.3.2 |
| VISION[118,119] | 2021 | $^{177}$Lu-PSMA-617 | SOC | III | 2 | 814 | 831 | Rx: 551<br>Ctrl: 280 | Rx: 70 (48-94)<br>Ctrl: 71.5 (40-89) | rPFS OS | Yes | 1.4.1 |
| PSMAfore[120,121] | 2023 | $^{177}$Lu-PSMA-617 | Enzalutamide/ abiraterone | III | 2 | 450 | 469 | Rx: 234<br>Ctrl: 234 | Rx: 71 (65-77)[a]<br>Ctrl: 72 (67-77)[a] | rPFS | Yes | 1.4.1 |

(continued on following page)

**TABLE 1.** Summary of Characteristics of Included Trials (continued)

| Trial | Year | Treatment Arm | Control Arm | Phase | Arms | Estimated Accrual, No. | Actual Accrual, No. | Participants, No. | Age, Years, Median (range) | Primary End Point | QoL Reported | Guideline Recommendation |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CARD[122,123] | 2019 | Cabazitaxel | Enzalutamide/ abiraterone | III | 2 | 234 | 255 | Rx: 129<br>Ctrl: 126 | Rx: 70 (46-85)<br>Ctrl: 71 (45-88) | rPFS | Yes | 1.4.1 |
| MDACC Study[124] | 2019 | Cabazitaxel + carboplatin | Cabazitaxel | II | 2 | 160 | 160 | Rx: 81<br>Ctrl: 79 | Rx: 68 (62-73)[a]<br>Ctrl: 66 (61-69)[a] | cPFS | No | 1.4.4<br>2.2 |
| FIRSTANA[125] | 2017 | Cabazitaxel[c] | Docetaxel | III | 3 | 1,170 | 1,168 | Arm1: 389<br>Arm2: 388<br>Ctrl: 391 | Arm1: 68 (44-90)<br>Arm2: 68.5(42-85)<br>Ctrl: 69 (41-87) | OS | Yes | 1.1.3 |
| TRITON-3[126] | 2023 | Rucaparib | Enzalutamide/ abiraterone/ docetaxel | III | 2 | 400 | 405 | Rx: 270<br>Ctrl: 135 | Rx: 70 (45-90)<br>Ctrl: 71 (47-92) | rPFS | Yes | 1.2.1<br>1.4.2 |

Abbreviations: cPFS, clinical progression-free survival; Ctrl, control; NA, not available; ORR, objective response rate; OS, overall survival; PSA, prostate-specific antigen; QoL, quality of life; rPFS, radiographic progression-free survival; Rx, treatment; SBRT, stereotactic body radiation therapy; SOC, standard of care; TTP, time to progression.

[a]Age is reported as median (IQR).

[b]Arm 1 received docetaxel 75 mg/$m^2$ once every 3 weeks; arm 2 received docetaxel 30 mg/$m^2$ once every week.

[c]Arm 1 received cabazitaxel 20 mg/$m^2$ once every 3 weeks; arm 2 received cabazitaxel 25 mg/$m^2$ once every 3 weeks.

PFS due to unblinded outcome assessment as shown in the Data Supplement (Fig S2: Risk of bias across the included trials in this guideline).

### Evidence Quality Assessment

Two independent reviewers assessed the risk of bias in the included trials using the Cochrane risk of bias assessment tool version 2. Certainty of evidence at the level of each outcome was adjudicated using the GRADE approach.[7] Direct evidence for each pairwise estimate was assessed for overall risk of bias, inconsistency, indirectness, imprecision, and potential publication bias. A noncontextualized approach with null effect as the threshold of importance was utilized for assessing imprecision. The responses were graded as high, moderate, low, or very low, and the results are presented as summary of findings tables for each question-specific recommendation.

### Evidence Summary

Detailed evidence summaries are provided as the Data Supplement (Table S4: Summary of findings tables by clinical question).

## RECOMMENDATIONS

All recommendations are available in Table 2.

## CLINICAL QUESTION 1.1

What are the treatment options for patients previously treated with ADT alone in castration-sensitive or nonmetastatic castration-resistant prostate cancer (CRPC) setting and whose disease has progressed to metastatic CRPC?

### Literature Review and Analysis

1.1.1. In patients with *BRCA1*/2 gene alterations, meta-analysis showed that the combination of poly(ADP-ribose) polymerase inhibitors (PARPi) and an ARPI was associated with statistically significant improvement in PFS and OS as compared to ARPI alone (PROpel,[78-84] TALAPRO-2,[73-77] MAGNITUDE[67-72]; PFS, HR, 0.28 [95% CI, 0.13 to 0.62]; OS, HR, 0.47 [95% CI, 0.31 to 0.71]). Results of individual trials are provided in the Data Supplement (Table S4: Summary of findings tables by clinical question).

1.1.2. Outcomes by other homologous recombinant repair (HRR) genes were inconsistently reported across the included trials. In patients with non-*BRCA1*/2 HRR gene alterations, the combination of talazoparib and enzalutamide was associated with statistically significant improvement in PFS but not in OS as compared to enzalutamide only (TALAPRO-2[73-77]; PFS, HR, 0.71 [95% CI, 0.52 to 0.96]; OS, HR, 0.66 [95% CI, 0.40 to 1.10]).

1.1.3. Combination of abiraterone with prednisone was associated with a statistically significant PFS and OS benefit compared to placebo in patients who received prior ADT alone (COU-AA-302[85-87]; PFS, HR, 0.52 [95% CI, 0.45 to 0.61]; OS, HR, 0.81 [95% CI, 0.70 to 0.93]). Adding stereotactic body radiotherapy (SBRT) to abiraterone and prednisone for patients with three or fewer metastases further improved PFS (HR, 0.35 [95% CI, 0.21 to 0.57]; $P < .001$), PSA response rate (>50%; odds ratio [OR], 5.3 [95% CI, 2.05 to 13.88]; $P = .001$), and complete PSA response rate (<0.2 ng/ml; OR, 4.3 [95% CI, 2.12 to 8.38]; $P < .001$).[13]

Enzalutamide was associated with a statistically significant PFS and OS benefit compared to placebo in patients who received prior ADT alone (PREVAIL[88-90]; PFS, HR, 0.32 [95% CI, 0.28 to 0.36]; OS, HR, 0.77 [95% CI, 0.67 to 0.88]).

Docetaxel was associated with a statistically significant OS benefit compared to mitoxantrone in patients who received prior ADT alone (TAX327[91,92]; OS, HR, 0.79 [95% CI, 0.67 to 0.93]). Docetaxel with estramustine was associated with an improved OS (HR, 0.80 [95% CI, 0.67 to 0.97]) compared to mitoxantrone with prednisone on SWOG 99-16 for patients treated with prior ADT.[14]

The recommendations supporting the utility of darolutamide or apalutamide as an alternative due to concerns of drug side effects and/or interactions are based on expert consensus and evidence from limited data.

In patients with allergic reactions or toxicity like peripheral neuropathy due to docetaxel, cabazitaxel is a suitable option according to expert consensus. There was no statistically significant difference in PFS or OS benefit seen in patients who received cabazitaxel compared with docetaxel (FIRSTANA[125]; PFS, HR, 0.96 [95% CI, 0.79 to 1.17]; OS, HR, 0.97 [95% CI, 0.82 to 1.16]). On FIRSTANA,[125] febrile neutropenia, diarrhea, and urinary symptoms were more common with cabazitaxel use, and peripheral neuropathy, hair and nail changes, and peripheral edema were more common with docetaxel use.

1.1.4. Radium 223 was associated with a statistically significant OS benefit compared to placebo in patients with symptomatic bone–only disease without known visceral metastases or large nodal (>3 cm in short axis) metastatic disease (ALSYMPCA[93,94]; OS, HR, 0.70 [95% CI, 0.58 to 0.83]).

1.1.5. Data from phase II KEYNOTE-158 trial[95,96] showed that patients (n = 6) with histologically and/or cytologically confirmed microsatellite instability-high (MSI-H) or mismatch repair-deficient who have disease progression on prior therapies exhibited a median PFS of 4.1 months (95% CI, 2.4 to 4.9) and a median OS of 23.5 months (95% CI, 13.5 to not reached) with pembrolizumab. Although this cohort had only six patients with MSI-H prostate cancer, this led to a histology-agnostic indication for MSI-H cancers treated with pembrolizumab.

1.1.6. Meta-analysis showed that sipuleucel-T was associated with a statistically significant OS benefit compared to

**TABLE 2.** Summary of All Recommendations

| Clinical Question | Recommendation |
|---|---|
| *General Note.* The following recommendations (strong or conditional/weak) and terminology [see the Data Supplement] represent reasonable options for patients depending on clinical circumstances and in the context of individual patient preferences. Recommended care should be accessible to patients whenever possible. | |
| Principles of practice | 1. mCRPC is defined as castrate levels of testosterone (<50 ng/mL or 1.7 nmol/L) with evidence of either new or progressive metastatic disease on radiologic assessment or two consecutive rising PSA levels (minimal start value is 1.0 ng/mL) in the setting of existing metastatic disease |
| | 2. The Panel recommends both germline and somatic testing for patients with metastatic prostate cancer at the earliest available opportunity. Additional recommendations and discussion regarding the same are available in the Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline[8] |
| | 3. The Panel recommends personalizing treatment based on prior therapies received during castration-sensitive setting, while also accounting for each patient's clinical status, cancer-related symptoms and signs, therapy-related toxicities, potential drug interactions, cost, and drug availability |
| | 4. The Panel recommends continuation of ADT (or surgical castration) to maintain castrate levels of testosterone for every patient with mCRPC |
| | 5. The Panel recommends early integration of palliative and supportive care teams for symptom management and to review goals of care for patients with mCRPC. Additional information on the same is available in the Palliative Care for Patients With Cancer: ASCO Guideline Update[9] |
| | 6. The Panel recommends the use of bone-protective agents like denosumab (RANK ligand inhibitor) or zoledronic acid (bisphosphonate) for patients with bone metastases to lower the risk of SREs. Additional recommendations on the same are available in the Bone Health and Bone-Targeted Therapies for Prostate Cancer: ASCO Endorsement of a Cancer Care Ontario Guideline[10] |
| | 7. Limited randomized clinical trial data exist for optimal sequencing of therapies for patients with mCRPC, as certain therapies studied in the mCRPC setting are now being utilized earlier during castration-sensitive disease |
| For patients previously treated with ADT alone in castration-sensitive or nonmetastatic CRPC setting and whose disease has progressed to metastatic CRPC, the panel recommends (see Fig 1: Algorithm for patients previously treated with ADT alone) | 1.1.1. The Panel recommends HRR testing prior to initiation of systemic therapy (see the Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline).[8] If positive for *BRCA1/2* alterations, then preferred options include either of niraparib + abiraterone, olaparib + abiraterone, or talazoparib + enzalutamide. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.1.1:* The Panel recommends personalized treatment based on therapies received during castration-sensitive setting, while also accounting for each patient's clinical status, therapy-related toxicities, potential drug interactions, cost, and drug availability as per Principles of Practice Recommendation 3 |
| | 1.1.2. The Panel suggests the combination of talazoparib and enzalutamide for patients with any of the other HRR alterations (*PALB2*, *ATM*, *ATR*, *CHEK2*, *FANCA*, *RAD51C*, *NBN*, *MLH1*, *MRE11A*, *CDK12*). (Evidence quality: Low, pending further follow up data; Strength of recommendation: Conditional) |
| | *Practical Information for Recommendation 1.1.2:* Responses vary according to individual genes, but the strongest responses are seen in patients with *PALB2* and *CDK12* alterations in post hoc subgroup analyses (Data Supplement, Table S5) |
| | 1.1.3. The Panel recommends single-agent therapy with either abiraterone with prednisone, enzalutamide, or chemotherapy utilizing docetaxel for patients without HRR alterations. (Evidence quality: High; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.1.3:* Although there are limited data for darolutamide or apalutamide in mCRPC setting, either agent can be used as an alternative to other ARPIs if there are concerns about side effects or drug interactions (expert opinion). Cabazitaxel is an option if patients have an allergic reaction or toxicity like peripheral neuropathy which precludes the use of docetaxel. The choice between an ARPI and chemotherapy may depend on the patient's health, their ability to tolerate chemotherapy, the presence of symptoms and/or visceral disease, accessibility, cost, and the patient's preference, especially if they are averse to chemotherapy |
| | 1.1.4. The Panel recommends radium 223 in patients with symptomatic bone–only disease without known visceral metastases or large nodal (>3 cm in short axis) metastatic disease. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.1.4:* Avoid use of radium 223 with abiraterone + prednisone due to the increased risk of fractures and mortality. Symptomatic disease is defined as consistent use of pain medications or undergoing external-beam radiation therapy for cancer-related bone pain in the last 12 weeks |
| | 1.1.5. The Panel recommends pembrolizumab for patients with MSI-high or dMMR after progression on other agents listed previously. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.1.5:* For patients with tumor mutational burden-high (TMB-H ≥10 mut/Mb), the Panel recommends considering other therapeutic options listed previously prior to utilization of pembrolizumab, given modest clinical benefit |
| | 1.1.6. The Panel suggests sipuleucel-T for selected patients with indolent disease course characterized by slow rising PSA, asymptomatic or minimally symptomatic disease, and low disease volume without visceral metastases. (Evidence quality: Moderate; Strength of recommendation: Conditional) |
| | *Practical Information for Recommendation 1.1.6:* The Panel recommends educating patients and caregivers regarding the lack of benefit in objective intermediate measurable end points (PSA levels, radiographic changes) associated with the use of sipuleucel-T |
| | 1.1.7. The Panel suggests metastasis-directed therapy in the form of radiation or surgical resection after a multidisciplinary evaluation in selected patients with oligometastatic progression. (Evidence quality: Low; Strength of recommendation: Conditional) |

(continued on following page)

**TABLE 2.** Summary of All Recommendations (continued)

| Clinical Question | Recommendation |
|---|---|
| For patients previously treated with ADT and an ARPI and whose disease has progressed to metastatic CRPC (see Fig 4: Algorithm for patients previously treated with ADT and an ARPI) | 1.2.1. The Panel recommends using olaparib monotherapy for patients with *BRCA1/2* alterations. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.2.1:* Rucaparib may be offered as an alternative to olaparib for selected patients based on clinical judgment and patient preferences, pending availability |
| | 1.2.2. The Panel suggests olaparib monotherapy for patients with any of the other HRR alterations. (Evidence quality: Very Low; Strength of recommendation: Conditional) |
| | *Practical Information for Recommendation 1.2.2:* Patients with *BRCA1*, *BRCA2*, *CDK12*, and *PALB2* gene alterations experienced the greatest benefit with PARP inhibitors. The data supporting continuation of ARPI with a PARP inhibitor are limited. A second ARPI can be considered based on clinical judgment, patient co-morbidities, and factors for slowly progressive disease determined by slowly rising PSA without any radiologic progression or cancer-related symptoms. The Panel suggests referring to Recommendation 1.2.2.1 for alternate options |
| | 1.2.2.1. The Panel recommends using docetaxel chemotherapy or radium 223, based on Recommendations 1.1.3 and 1.1.4, respectively, for patients without HRR alterations. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.2.2.1:* Cabazitaxel may be considered for selected patients if they have an allergic reaction or toxicity like peripheral neuropathy which precludes the use of docetaxel |
| | 1.2.3. The Panel suggests sipuleucel-T or local therapies per Recommendations 1.1.6 or 1.1.7, respectively, for selected patients. (Evidence quality: Moderate; Strength of recommendation: Conditional) |
| For patients previously treated with ADT and docetaxel, and whose disease has progressed to metastatic CRPC (see Fig 2: Algorithm for patients previously treated with ADT and docetaxel) | 1.3.1. The Panel recommends following Recommendations 1.1.1 and 1.1.2 for patients with HRR alterations. (Evidence quality: Moderate to Low; Strength of recommendation: Strong) |
| | 1.3.2. The Panel recommends either an ARPI such as abiraterone with prednisone or enzalutamide, or chemotherapy with cabazitaxel for patients without HRR alterations. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.3.2:* The choice between an ARPI or chemotherapy in this situation depends on the patient's clinical status, presence or absence of symptoms from metastatic disease, cost and drug availability, and ability to tolerate therapy. Docetaxel rechallenge may be offered to patients with durable response to docetaxel (patients who received initial docetaxel therapy and achieved a PSA of <0.2 after 8 months of treatment, with no evidence of clinical or radiologic progression) in the CSPC setting |
| | 1.3.3. The Panel also recommends the use of radium 223 or pembrolizumab in this setting (as per Recommendation 1.1.4 and 1.1.5) (Evidence quality: Moderate [for both radium 223 and pembrolizumab]; Strength of recommendation: Strong) |
| For patients previously treated with ADT, ARPI, and docetaxel, and have progressive metastatic CRPC (see Fig 3: Algorithm for patients previously treated with ADT, ARPI and docetaxel) | 1.4.1. The Panel recommends $^{177}$Lu-PSMA-617 for PSMA-positive disease or chemotherapy using cabazitaxel. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.4.1:* For patients with PSMA-negative disease, chemotherapy with cabazitaxel or radium 223 (Recommendation 1.1.4) for symptomatic bone-predominant metastatic disease may be offered |
| | 1.4.2. The Panel recommends olaparib monotherapy for patients with HRR alterations. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 1.4.2:* As it has demonstrated a PFS benefit, rucaparib may be offered as an alternative to olaparib for selected patients with *BRCA1/2* alterations based on clinical judgment and patient preferences, pending availability |
| | 1.4.3. The Panel also recommends radium 223 or pembrolizumab for specific situations outlined earlier (as per Recommendations 1.1.4 and 1.1.5). (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | 1.4.4. The Panel suggests clinical trials, cabazitaxel + carboplatin, carboplatin monotherapy, best supportive care, and hospice care for patients with disease progression despite previously mentioned options. (Evidence quality: Low [cabazitaxel + carboplatin]; Low [carboplatin monotherapy]; NA [best supportive care and hospice care]; Strength of recommendation: Conditional) |
| What are the treatment options for de novo or treatment-emergent small cell neuroendocrine carcinoma of the prostate? (see Fig 5: Algorithm for treatment options for de novo or treatment-emergent small cell neuroendocrine carcinoma of the prostate) | 2.1. The Panel recommends utilizing cisplatin or carboplatin plus etoposide for first-line systemic therapy as per *Systemic Therapy for Small-Cell Lung Cancer: ASCO-Ontario Health (Cancer Care Ontario) Guideline.*[11] (Evidence quality: Low; Strength of recommendation: Strong) |
| | 2.2. The Panel suggests carboplatin plus cabazitaxel for patients with small cell neuroendocrine prostate cancer or those with deleterious alterations in two or more of *TP53*, *RB1*, and *PTEN* genes. (Evidence quality: Low; Strength of recommendation: Conditional) |
| | 2.3. The Panel recommends enrollment in clinical trials after progression on platinum-based chemotherapy for eligible patients. (Evidence quality: NA; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 2.3:* Broad NGS testing at the time of disease progression may aid assessment of clinical trial eligibility |
| | 2.3.1. In selected patients, the Panel suggests upfront chemotherapy (carboplatin and/or cisplatin plus etoposide) with immunotherapy followed by maintenance immunotherapy, lurbinectedin, topotecan, and tarlatamab may be utilized based on extrapolation from studies for small cell lung cancer. Prospective clinical studies for these agents for small cell prostate cancer are lacking and use of these agents should be considered on a case-by-case basis only. (Evidence quality: Very Low; Strength of recommendation: Conditional) |
| | *Practical Information for Recommendation 2.3.1:* Clinicians should consider the possibility of treatment-emergent small cell or neuroendocrine carcinoma or an aggressive variant prostate cancer in patients with exclusively predominant visceral metastases with low PSA, predominant lytic bone lesions, bulky (≥5 cm) lymphadenopathy, rapid disease progression within 6 months of initiation of hormonal therapy, and alterations in any two of the three tumor suppressor genes *TP53*, *RB1*, and *PTEN*. |

(continued on following page)

**TABLE 2.** Summary of All Recommendations (continued)

| Clinical Question | Recommendation |
|---|---|
| How to assess for response while on systemic therapy for mCRPC? | 3.1. The Panel recommends using a combination of 1) clinical assessments to evaluate the tolerability of cancer therapy and cancer-related symptoms, 2) blood tests including PSA, CBC with differential, and CMP, and 3) radiologic assessments to determine if patients are benefiting from systemic therapy. (Evidence quality: NA; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 3.1:* The use of rising PSA alone to determine disease progression in mCRPC, without evidence of clinical or radiographic progression is not recommended |
| What scans are recommended for response assessment? | 3.2.1. The Panel recommends use of Tc99 bone scan and CT CAP to define progression for patients with mCRPC. These radiologic assessments can be performed every 3-6 months or in patients with rising PSA levels and/or evidence of clinical progression. (Evidence quality: Moderate; Strength of recommendation: Strong) |
| | *Practical Information for Recommendation 3.2.1:* The Panel suggests PCWG3[12] criteria be referred to in the clinical decision-making process. Currently, the role of PSMA PET-CT is limited to identify PSMA-expressing tumors for patients who will benefit with $^{177}$Lu-PSMA-617. In selected patients, PSMA PET can be used to stage patients with a rising PSA and concern for progression not visualized on conventional imaging. Prospective data on using PSMA PET scans for response assessment are still emerging, and at this time, the Panel does not recommend their routine use |

NOTE. The strength of the recommendation is defined as follows. Strong: In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention. Conditional/weak: In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.

Abbreviations: $^{177}$Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen-deprivation therapy; ARPI, androgen receptor pathway inhibitor; CAP, chest, abdomen, and pelvis; CMP, comprehensive metabolic panel; CRPC, castration-resistant prostate cancer; CSPC, castration-sensitive prostate cancer; CT, computed tomography; dMMR, mismatch repair-deficient; HRR, homologous recombinant repair; mCRPC, metastatic castration-resistant prostate cancer; MSI-high, microsatellite instability-high; NA, not applicable; NGS, next-generation sequencing; PARP, poly(ADP-ribose) polymerase; PCWG, Prostate Cancer Working Group; PET, positron emission tomography; PFS, progression-free survival; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen; RANK, receptor activator of nuclear factor-kappa B; SRE, skeletal-related event; TMB-H, tumor mutational burden-high.



placebo in patients with an indolent disease course with slow rising PSA, and asymptomatic or minimally symptomatic disease without visceral metastatic disease (IMPACT,[99] D9902A,[98] D9901[97,98]; OS, HR, 0.73 [95% CI, 0.61 to 0.98]).

1.1.7. Data from the phase II ARTO trial[100] showed that the addition of SBRT to abiraterone acetate showed statistically significant improvement in PFS (HR, 0.35 [95% CI, 0.21 to 0.57]) in patients with oligometastatic mCRPC.

## CLINICAL QUESTION 1.2

What are the treatment options for patients previously treated with ADT and an ARPI and whose disease has progressed to metastatic CRPC?

### Literature Review and Analysis

1.2.1. In patients with *BRCA1/2* gene alterations, olaparib was associated with statistically significant improvement in PFS and OS as compared to enzalutamide or abiraterone (PROfound[101-106]; PFS, HR, 0.22 [95% CI, 0.15 to 0.32]; OS, HR, 0.63 [95% CI, 0.42 to 0.95]).

The patients who received prior ARPIs in PROpel,[78-84] TALAPRO-2,[73-77] and MAGNITUDE[67-72] are <10% of the total population.

In patients with *BRCA1/2* gene alterations, rucaparib monotherapy was associated with statistically significant improvement in PFS but not in OS as compared to physician's choice control (TRITON-3[126]; PFS, HR, 0.50 [95% CI, 0.36 to 0.69]; OS, HR, 0.81 [95% CI, 0.58 to 1.21]).

1.2.2. In patients with HRR gene alterations, olaparib monotherapy was associated with statistically significant improvement in PFS but not in OS compared to enzalutamide or abiraterone (PROfound[101-106]; PFS, HR, 0.49 [95% CI, 0.38 to 0.63]; OS, HR, 0.79 [95% CI, 0.61 to 1.03]). However, fixed-effect meta-analysis pooling data for non-*BRCA1/2* HRR genes showed no statistically significant improvement in PFS (HR, 0.92 [95% CI, 0.64 to 1.03]) or in OS (HR, 1.03 [95% CI, 0.70 to 1.50]).

1.2.2.1. The results for docetaxel (TAX327,[91,92] SWOG 99-16[127]) and radium 223 (ALSYMPCA[93,94]) are described in literature review and analysis subsections 1.1.3 and 1.1.4, respectively.

1.2.3. The meta-analyzed estimate for sipuleucel-T is described in literature review and analysis subsection 1.1.6. The results for addition of local therapies are described in literature review and analysis subsection 1.1.7.

## CLINICAL QUESTION 1.3

What are the treatment options for patients previously treated with ADT and docetaxel, and whose disease has progressed to metastatic CRPC?

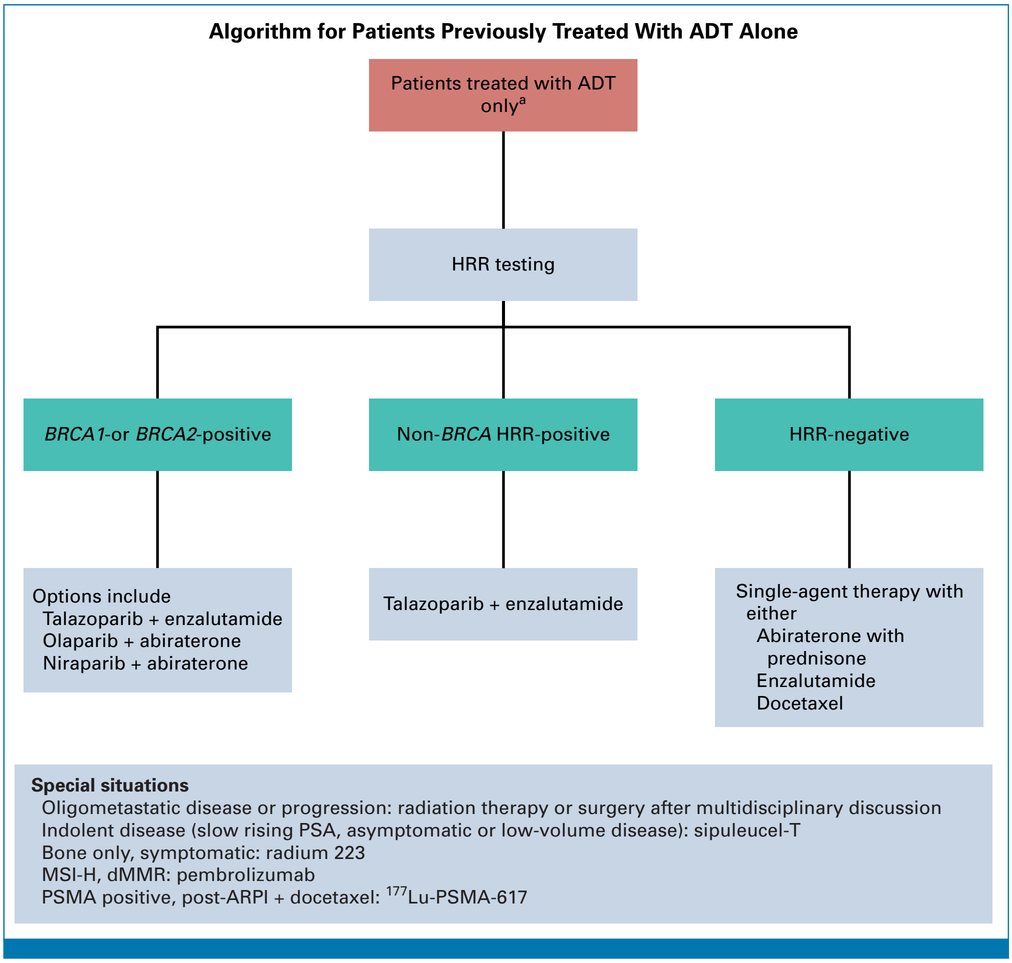


**FIG 1.** Algorithm for patients previously treated with ADT alone. [a]Options listed are preferred options. $^{177}$Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen-deprivation therapy; ARPI, androgen receptor pathway inhibitor; dMMR, mismatch repair-deficient; HRR, homologous recombinant repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen.



### Literature Review and Analysis

1.3.1. The results for the combination of PARP inhibitors and ARPIs in patients with *HRR* gene alterations are described in 1.1.1 and 1.1.2, respectively. Results of individual trials are provided in the Data Supplement (Table S4: Summary of findings tables by clinical question).

1.3.2. The treatment combination of abiraterone with prednisone was associated with a statistically significant PFS and OS benefit compared to placebo in patients who had received prior treatment with ADT and docetaxel (COU-AA-301[107-112]; PFS, HR, 0.66 [95% CI, 0.58 to 0.76]; OS, HR, 0.74 [95% CI, 0.64 to 0.86]).

Enzalutamide was associated with a statistically significant PFS and OS benefit when compared with placebo in patients who had received prior treatment with ADT and docetaxel (AFFIRM[113-115]; PFS, HR, 0.40 [95% CI, 0.35 to 0.47]; OS, HR, 0.63 [95% CI, 0.53 to 0.75]).

Cabazitaxel was associated with a statistically significant PFS and OS benefit as compared to mitoxantrone in patients who had received prior treatment with ADT and docetaxel (TROPIC[116,117]; PFS, HR, 0.74 [95% CI, 0.64 to 0.86]; OS, HR, 0.70 [95% CI, 0.59 to 0.83]).

1.3.3. The results for radium 223 (ALSYMPCA[93,94]) and pembrolizumab (KEYNOTE-158[95,96]) are described in literature review and analysis subsections 1.1.4 and 1.1.5, respectively.

## CLINICAL QUESTION 1.4

What are the treatment options for patients previously treated with triplet therapy (ADT, ARPI, and docetaxel), and whose disease has progressed to metastatic CRPC?

### Literature Review and Analysis

1.4.1. Lutetium-177-PSMA-617 ($^{177}$Lu-PSMA-617) was associated with a statistically significant PFS and OS

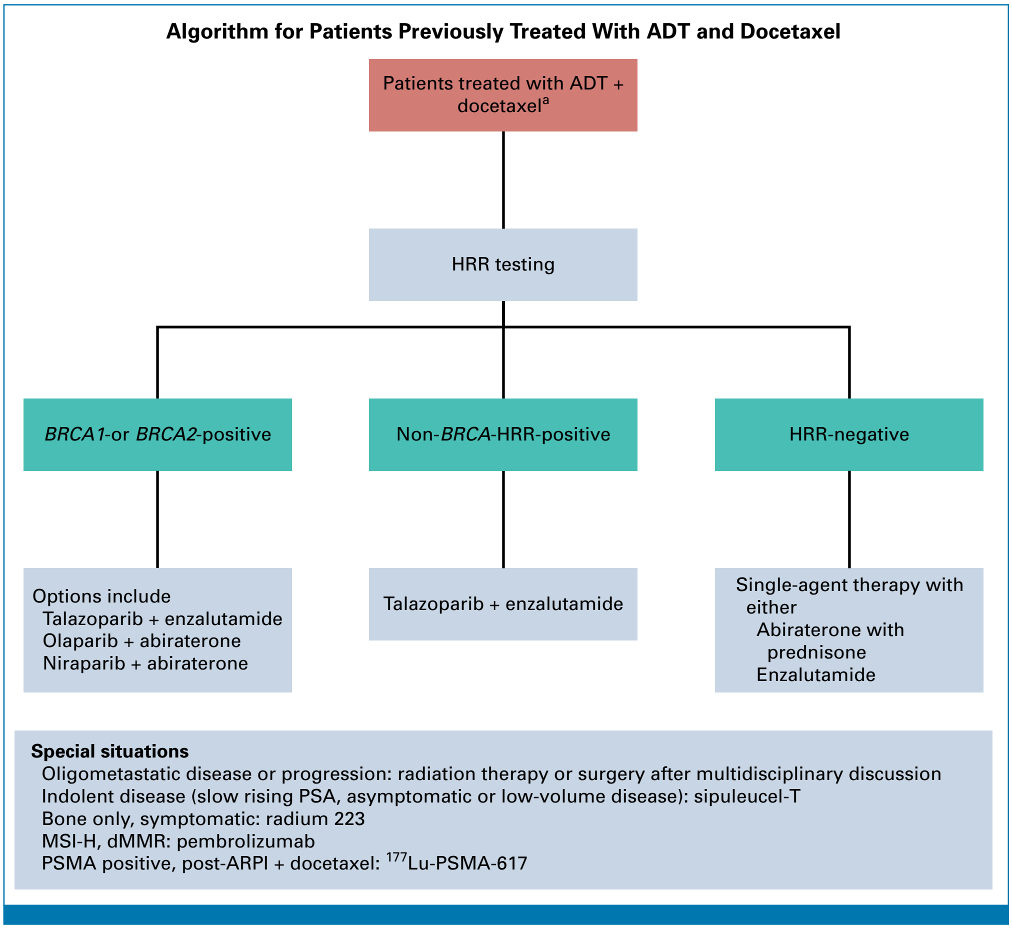


**FIG 2.** Algorithm for patients previously treated with ADT and docetaxel. [a]Options listed are preferred options. $^{177}$Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen-deprivation therapy; ARPI, androgen receptor pathway inhibitor; dMMR, mismatch repair-deficient; HRR, homologous recombinant repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen.

benefit compared to the standard of care (VISION[118,119]; PFS, HR, 0.40 [95% CI, 0.28 to 0.56]; OS, HR, 0.62 [95% CI, 0.52 to 0.74]).

Cabazitaxel was associated with a statistically significant PFS and OS benefit compared to enzalutamide + abiraterone (CARD[122,123]; PFS, HR, 0.54 [95% CI, 0.40 to 0.73]; OS, HR, 0.64 [95% CI, 0.46 to 0.89]).

1.4.2. In patients with HRR gene alterations, olaparib monotherapy was associated with statistically significant improvement in PFS but not OS as compared to enzalutamide or abiraterone (PROfound[101-106]; PFS, HR, 0.49 [95% CI, 0.38 to 0.63]; OS, HR, 0.79 [95% CI, 0.61 to 1.03]).

The results of rucaparib monotherapy from the TRITON-3 trial[126] are described in the literature review and analysis subsection 1.2.1.

1.4.3. The results for radium 223 (ALSYMPCA[93,94]) and pembrolizumab (KEYNOTE-158[95,96]) are described in the literature review and analysis subsections 1.1.4 and 1.1.5, respectively.

1.4.4. The treatment combination of cabazitaxel and carboplatin was associated with a statistically significant PFS benefit compared to cabazitaxel alone in patients in heavily pretreated mCRPC patients (phase I/II MDACC study[124]; PFS, HR, 0.69 [95% CI, 0.50 to 0.95]).

The recommendations supporting the utility of carboplatin monotherapy, best supportive care, and palliative care are based on expert consensus, and evidence from control arms of the included trials.

## DISCUSSION

The treatment of metastatic castration-sensitive prostate cancer (mCSPC) has evolved rapidly in the past few years. In a pivotal large phase III clinical study with 3,040 patients, intermittent ADT was not noninferior to continuous ADT for treatment of mCSPC.[15] Thus continuous ADT became the mainstay of treating patients with mCSPC for a long period of time. Subsequent advances led to approval of doublet regimens ADT with docetaxel[16] and ADT with abiraterone and prednisone[17] for patients with high-

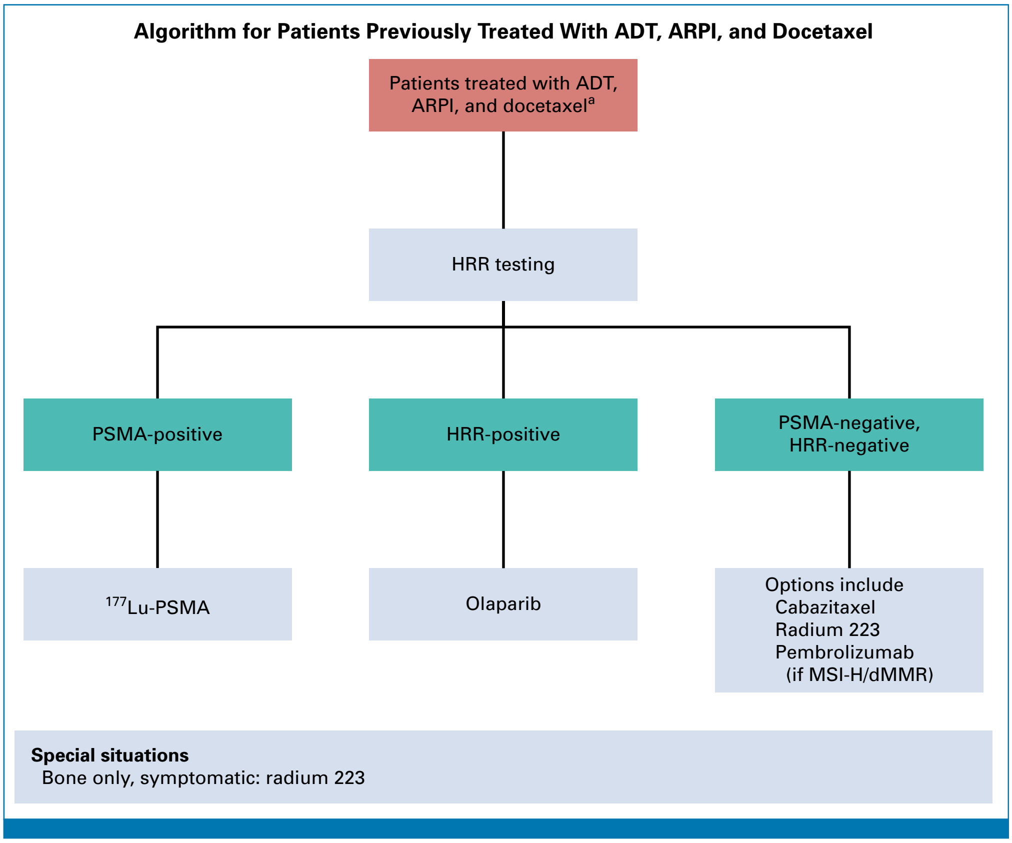


**FIG 3.** Algorithm for patients previously treated with ADT, ARPI, and docetaxel. [a]Options listed are preferred options. $^{177}$Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen-deprivation therapy; ARPI, androgen receptor pathway inhibitor; dMMR, mismatch repair-deficient; HRR, homologous recombinant repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen.



volume mCSPC. The use of doublet therapy subsequently expanded to all patients with mHSPC with ADT with enzalutamide[18] and ADT with apalutamide.[19] More recently, two large phase III clinical trials supported the use of triplet therapy ADT with darolutamide and docetaxel[20] and ADT with abiraterone, prednisone, and docetaxel[17] for patients with mCSPC. Thus, treatment for mCSPC has evolved and intensified rapidly from single-agent ADT with most patients currently receiving doublet and triplet regimens. Our current guidelines take into consideration this heterogeneity in frontline therapy for mCSPC for specific recommendations for their treatments for mCRPC.

Treatments options include ARPI, chemotherapy, PARP inhibitors, radiopharmaceuticals, and immunotherapy for select patients. Moreover, the therapeutic agents that were first developed and utilized for mCRPC such as abiraterone, enzalutamide, and docetaxel are now utilized in the castration-sensitive setting, leading to a gap in the guidance of treatment sequencing in mCRPC setting. The eligibility criteria for the clinical trials that led to the development of these therapeutic agents vary widely, making it challenging to determine the optimal sequencing and combination of these agents for identifying the most effective treatment for each patient.

Given that numerous real-world studies show significant attrition of patients with each line of therapy (LOT), it is essential to use therapies with improved survival at the right time.[21-23] Additionally, early biomarker testing is crucial before starting treatment for patients with mCRPC. HRR gene alterations are common in mCRPC.[24,25] Patients with *BRCA1/2* alterations have significant survival benefit with PARPi and ARPI combinations, and hence it is important to test for these biomarkers and optimize the first-line therapy for mCRPC. It is also important to note that the survival benefit is most significant for patients with *BRCA1/2* when compared to non-*BRCA1/2 HRR* alterations. The combination of PARPi and ARPI therapies is most effective for patients who have not previously received ARPI. It is also important to note that in the BRCAAway study,[128] PFS benefit was significantly better with concurrent use of olaparib and

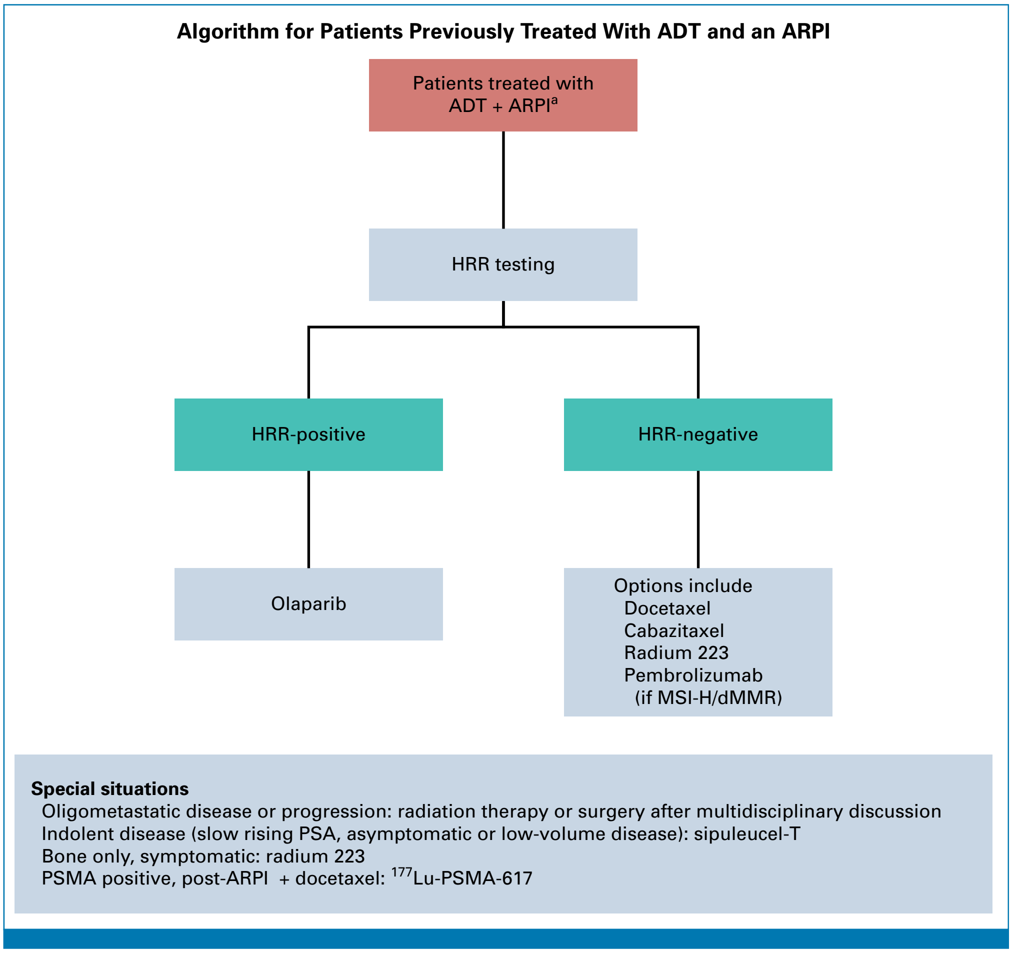


**FIG 4.** Algorithm for patients previously treated with ADT and an ARPI. [a]Options listed are preferred options. $^{177}$Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen-deprivation therapy; ARPI, androgen receptor pathway inhibitor; dMMR, mismatch repair deficient; HRR, homologous recombinant repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen.



abiraterone in patients with *BRCA1*/2 or *ATM* mutations when compared to either of drugs used alone or in sequence.[26]

There remain limited data for sequencing treatments in patients with mCRPC. In a RCT involving patients with mCRPC, abiraterone followed by enzalutamide showed better clinical efficacy compared to the reverse sequence of enzalutamide followed by abiraterone.[27] However, sequential use of ARPI agents has typically resulted in lower response rates with the second ARPI. In the CARD[122,123] RCT, cabazitaxel when compared to a second ARPI (enzalutamide or abiraterone with prednisone) in patients with mCRPC showed improved PFS and OS (13.6 *v* 11.0 months; HR, 0.64 [95% CI, 0.46 to 0.89]; *P* = .008).[28] Nevertheless, real-world evaluations of treatment patterns indicate that second-line ARPI use remains more common.[21,29] Factors such as patient preference, concerns over side effects and worsening quality of life with chemotherapy, advanced age, frailty, and multiple comorbidities often contribute to these treatment patterns.

$^{177}$Lu-PSMA-617, a PSMA-directed radiopharmaceutical therapy is a newer treatment option for patients with mCRPC with PSMA-positive imaging and previously treated with an ARPI and chemotherapy.[30] The recent clinical trials of $^{177}$Lu-PSMA-617 in the PSMAfore study[31] and $^{177}$Lu-PNT2002[30] in the SPLASH study[129] have provided insights for the role of radiopharmaceuticals prior to chemotherapy. In the PSMAfore study, $^{177}$Lu-PSMA-617 administered at the dose of 7.4 GBq once every 6 weeks for six cycles, when compared to the arm with an alternate ARPI, showed significantly prolonged median rPFS (11.60 *v* 5.59 months) with a HR of 0.49 (95% CI, 0.39 to 0.61). There was no OS difference between the two groups at the time of analysis; however, about 51% of patients in the ARPI arm crossed over to receive $^{177}$Lu-PSMA-617, which could potentially impact the OS. Similarly, in the SPLASH study,[129] the efficacy of another radiopharmaceutical, $^{177}$Lu-PNT2002, was compared to ARPI switch therapy. In this study, 177Lu-PNT2002 was administered at a slightly lower dose of 6.8 GBq once every 8 weeks and for up to four cycles. In this study as well, there was rPFS benefit without an OS benefit, possibly due to crossover. Both studies reported

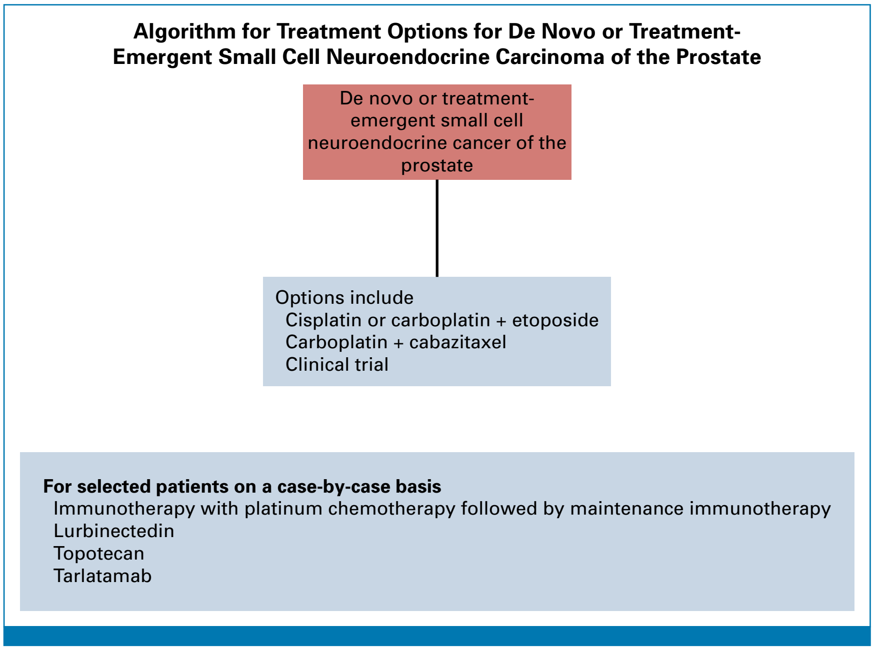


**FIG 5.** Algorithm for treatment options for de novo or treatment-emergent small cell neuroendocrine carcinoma of the prostate.

improved quality of life for patients with the radiopharmaceutical agents. The Panel recognizes the potential advantages of 177Lu-based therapies in the prechemotherapy setting for mCRPC. However, the Panel will follow-up on additional data and US Food and Drug Administration (FDA) review of both studies to inform future recommendations regarding the integration of radiopharmaceuticals prior to chemotherapy. The Panel is aware of initial analysis of the PEACE-3 trial[130] combining radium 223 to enzalutamide in patients with mCRPC. The combination was associated with improved rPFS and OS compared to enzalutamide alone. Side-effect profile for the combination of radium-223 and enzalutamide was excellent. The Panel will follow-up on additional recommendations once final OS analysis, additional data, and FDA review are complete.

The role of metastasis-directed therapy (MDT; either SBRT or surgery) for patients with mCRPC is formally outside of the scope of this guideline. However, there is a small randomized trial showing that patients with oligoprogression to mCRPC benefit from abiraterone and prednisone and SBRT. Another potential benefit of MDT is that if a patient experiences oligoprogression while on a potent systemic therapy (or combination), patients may be able to stay on that therapy longer while those oligoprogressive lesions are treated. More research is needed in this area, particularly in the use of positron emission tomography compared with conventional imaging to guide SBRT. Radiotherapy and surgery are important for palliative treatment of metastases due to pain, bleeding, obstruction, or cord compression.

A subset of patients with prostate cancer will develop a transformation to neuroendocrine prostate cancer (NEPC). Pure small cell carcinoma is an even rarer malignancy seen in <1% of patients with prostate cancer. Most commonly, NEPC arises in the setting of mCRPC with development of visceral metastatic disease, lytic bony metastatic lesions, very low or undetectable PSA levels, and rapid clinical progression. This is often classified as treatment-emergent NEPC (tNEPC). Loss of the *TP53*, *RB1*, and *PTEN* genes and alterations of the DNA repair genes like *BRCA2*, *BRCA1*, and *ATM* are commonly associated with development of tNEPC and aggressive variant prostate cancer (AVPC).[32] Frontline therapy for NEPC and/or small cell carcinoma of the prostate generally involves the use of cisplatin or carboplatin with etoposide. The use of these regimens and overall management has been extrapolated from treatment of metastatic small cell lung cancer due to lack of large prospective clinical trials for NEPC. In a multi-institutional retrospective study with 41 patients with advanced unresectable or metastatic NEPC, the use of platinum-based combination chemotherapy was associated with a median OS of 10 months. Patients with locally advanced disease had better prognosis compared to metastatic disease.[33] The use of cabazitaxel with carboplatin was evaluated in an open-label phase I-II clinical study in patients with AVPC. Combination carboplatin-cabazitaxel had improved median PFS of 7.3 months versus 4.5 months for cabazitaxel. However, the combination was also associated with a higher risk of fatigue and cytopenias.[34] In a small retrospective analysis, lurbinectedin use for NEPC was associated with a clinical benefit rate of 56%, OS of 6.1 months, and PFS of 3.35 months.[35] In another multi-institutional retrospective study, multiple regimens (taxanes, platinum, immunotherapy, and other systemic therapies) were used for second line-treatment of NEPC with an overall response rate of 23.5% and median OS of 7.4

months.[36] The use of multiple regimens without a clear standard of care reflects a clear lack of consensus for second-line treatment of NEPC. Due to lack of randomized clinical studies, the Panel suggests enrollment on a clinical study for treatment of NEPC in the second-line setting and beyond.

## LIMITATION OF THE RESEARCH AND FUTURE RESEARCH

This guideline is limited by a small number of clinical trials and study-level data informing current recommendations. Given the heterogeneous inclusion criteria across the included trials, meta-analysis for each treatment recommendation was not performed. An individual patient data meta-analysis pooling data from multiple trials adjusting for potential covariates can offer further insights. The reporting of cancer-specific outcomes was not consistent across the included trials, which precluded meaningful statistics. Not all trials reported rPFS; hence, time to tumor progression and composite PFS (a composite of radiographic disease and/or PSA progression) were used as surrogate end points to PFS. Moreover, it may be argued that rPFS may be an advantageous end point for some therapies like ARPIs that are administered indefinitely until disease progression as opposed to cyclic treatments like taxane-based chemotherapy, which are administered for a fixed number of cycles in the trials. However, the Expert Panel has also assessed OS benefit, which is a more robust end point and found a consistent pattern of results. It is also important to note that the receipt of subsequent life-prolonging therapies and crossover across the included trials could have underestimated OS and data adjusted for crossover and subsequent therapies were not reported in most trials. Likewise, there was considerable inconsistency and heterogeneity in the reporting of efficacy by different subgroups. Hence, formalized subgroup analyses were not performed to assess treatment effect modification.

The standard of care for advanced prostate cancer has evolved rapidly over the past decade, leading to challenges in the applicability of existing trial data. Most clinical trials used to formulate this guideline were conducted in an era when the current treatment options for hormone-sensitive disease were not available and do not reflect contemporary treatment regimens. As the Expert Panel developed guideline questions, they recognized the increasing use of intensified upfront treatments in mCSPC. However, this has resulted in a degree of indirectness in some recommendations, particularly related to patients with disease progression on triplet therapy in the mCSPC setting. The trials supporting these recommendations often lack patients whose disease previously progressed on triplet therapy in mCSPC, leading to a lower certainty of evidence. Consequently, clinicians should exercise caution when applying the data to patients who may not closely resemble those studied in earlier trials. The Expert Panel only included phase III trials and select phase I/II trials to inform the recommendations in the current guideline.

A number of novel therapies in the form of radioligand therapies, proteolysis-targeting chimera androgen receptor degraders, antibody-drug conjugates, chimeric antigen receptor-T cells, and bispecific T-cell engagers are being evaluated in clinical trials for treatment of mCRPC. Evidence in this space is rapidly evolving, and the living interactive evidence synthesis framework will be utilized to incorporate new data into the existing body of evidence as soon as they become available.

## PATIENT AND CLINICIAN COMMUNICATION

The following items were identified by the two patient representatives on the panel as additional factors that both patients and clinicians should consider and address throughout the emotional and medical journeys of someone facing CRPC:

- This patient group faces daunting survival odds. CRPC is a burdensome and life-threatening illness. Recent and ongoing innovation in cancer therapeutics, as outlined in these recommendations, offers patients increasing longevity, and as greater survival becomes possible, the quality of life lived during that time becomes increasingly important as well. Palliative care, medical care focused on providing relief from the symptoms and stress of the illness, can provide important benefits concurrent with or beyond the administration of cancer-directed therapy. Palliative care can improve the quality of life for patients and their families through early identification and treatment of pain, comorbid conditions, symptom control, and other challenges, whether they be physical, psychosocial, or spiritual. For these reasons, palliative care should begin early in the treatment process and be maintained throughout the entire treatment plan while responding to the changing needs of the patient.[37]
- This guideline represents diligent and exhaustive work by leading peers to address current issues and controversies of mCRPC treatment as supported by published data. The goal has been to assist clinicians facing enormous workloads and little time with insights on the latest data to help inform their treatment decisions.
- This patient community has substantially benefitted from the many developments addressed in this guideline. During the past 15 years, there has been incredible progress for patients because of their active participation in clinical trials that helped make discoveries and new treatments possible, and eligible patients should be encouraged to participate. However, these developments and the rate of change place a burden on clinicians and patients to recognize the rapidly changing therapeutic options. Therapy choice and sequencing are major challenges.
- Patients with CRPC need substantially greater information about the benefits and burdens of each new treatment that is being proposed.
- The choices on the sequencing of different treatments and what side effects to accept create significant confusion for patients with CRPC. Patients and physicians should work together to incorporate more second opinions in complex and challenging situations. Case conferences or tumor

boards should focus on collaborating on difficult cases to the greatest extent possible.
- The level of uncertainty associated with CRPC treatment is high, so this would indicate a crucial need for shared decision making between clinicians and patients. Additionally, the care of patients with CRPC requires input from multidisciplinary team members.
- Patients and caregivers should be encouraged to reach out to family members, friends, and other patients to reduce the loneliness, depression, and poor decision making that frequently occur by those attempting to go it alone.
- There is great need to improve personalized treatment for patients with CRPC. Substantial progress requires better establishing patient criteria for each of the recommended treatments discussed before. Identifying those who have the most efficacious outcomes for each treatment and the most tolerable side effects can potentially improve survival, reduce patient cost, and increase medical institution effectiveness. Understanding patients experiencing negative results is also critically important.

For recommendations and strategies to optimize patient clinician communication, see Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.[38]

## HEALTH EQUITY CONSIDERATIONS

Prostate cancer is one of the most common malignancies worldwide, with significant implications for all public health systems. As the global burden of prostate cancer continues to rise, and is expected to double until 2040,[39] it is important to understand that health disparities and inequities exist both on a global scale and within specific populations in the United States. These disparities are influenced by several factors, including socioeconomic status, access to health care, cultural particularities, and genetic predispositions, which will affect outcomes in prevention, diagnosis, treatment, and survivorship.

Social determinants of health, defined by the World Health Organization as the conditions in which an individual is born, grows, lives, works, and ages, can undermine ASCO's expert recommendations on best practices for prevention, screening, palliative and supportive care, and disease management for many patients with cancer.[40] On a global perspective, prostate cancer incidence and mortality rates vary significantly between regions. According to the Global Cancer Observatory,[41] high-income countries report the highest incidence rates, particularly in North America and Western Europe, while lower incidence rates are observed in parts of Asia and Africa. However, mortality rates often do not correlate directly with incidence; for instance, while African nations may report lower incidence rates, the mortality rates from prostate cancer are disproportionately high. It is expected that the major increase in incidence in the next decades, in fact, will come from low-middle–income countries,[39] which again imposes a huge challenge for patients and health services.

It is important to acknowledge that many people in the United States, as in other high-income countries, do not receive the highest level of cancer care due to the long-term impact of structural racism and the consequential unequal distribution of wealth among racial and ethnic groups, or people living in rural areas.[42]

In the United States, prostate cancer health disparities manifest closely related to racial, ethnic, and socioeconomic differences. African American patients are known to have the highest incidence and mortality rates of prostate cancer compared to other racial groups. Studies indicate that African American patients are more likely to be diagnosed at advanced stages of the disease and are also less likely to receive timely and adequate treatment.[43] Several prior studies have shown disparate enrollment in prostate cancer clinical trials.[44] As an example of these disparities, an extensive review focusing on minorities participation in prostate cancer clinical trials analyzed 72 global phase III and IV prevention, screening, and treatment prostate cancer clinical trials with enrollment start dates between 1987 and 2016, representing a total of 893,378 participants. Of the trials analyzed, 59 (81.9%) had available race data, and 11 (15.3%) of these trials additionally reported ethnicity. Participants were in the majority White patients (with the highest proportion in US nonpublicly funded trials), comprising over 96% of the study population. The proportion of White participants in prostate cancer clinical trials has remained at over 80% since 1990. Geographically, Africa and the Caribbean were particularly underrepresented with only 3% of countries included.[45] Therefore, real-world evidence or trials dedicated to these populations should be undertaken to evaluate treatment effect in underrepresented patient populations.

In the United States, it is still clear that many patients remain unable to reap the benefits of innovative prevention and early detection programs, biomarker testing, and new cancer therapies due to structural barriers including lack of transportation, stable housing, and adequate insurance coverage as well as food insecurity, health literacy, proximity to a dedicated cancer center, and cost of treatment and other services.[46] Additionally, sexual and gender minority people experience stigma along with barriers to cancer screening, prevention, and treatment that contribute to these cancer disparities.[47] Disparities widen in those who are also from a racial or ethnic minority, underscoring the influence of intersectionality in cancer health disparities.[48]

Further, geographic disparities can also impact the quality of care patients receive. Patients in rural settings are more likely to have worse survivorship outcomes and experience higher mortality rates compared to patients in nonrural settings. This can be attributed, in part, to a lower density of specialists and dedicated cancer centers, as only 21% of nonmetropolitan counties in the United States have one or more practicing oncologists.[49]

Another recently published example of these disparities studied newly diagnosed patients with mCRPC identified in Medicare fee-for-service claims during January 1, 2014-June 30, 2019. Among 14,780 patients with mCRPC, 22% did not receive life-prolonging therapy after their diagnosis, while 78% underwent at least one LOT. Of these, 42% received two or more LOTs, and 20% underwent three or more LOTs. Among patients receiving two or more LOTs, the most frequent first- to second-line treatment sequences were novel hormonal therapy (NHT) followed by a different NHT (33%), chemotherapy followed by NHT (14%), and NHT followed by chemotherapy (13%). Notably, only 28% of patients who started with NHT as their first-line therapy transitioned to a different therapy class for subsequent treatment. The median survival was 25.6 months after an mCRPC diagnosis and 23.4 months following the initiation of treatment. In summary, over one in five Medicare patients with mCRPC did not receive any life-prolonging therapy, and fewer than half underwent second-line treatment. NHTs were the most frequently used first- and second-line therapies, with sequential NHTs being the most common treatment strategy. These findings emphasize significant undertreatment of patients with mCRPC, particularly for therapies beyond NHTs, and underscore the prevalent use of sequential NHTs in real-world settings using real-world data.[21]

Several factors may contribute to these disparities:

1. Genetic and biological factors: Genetic predispositions can play a significant role in the higher aggressiveness of prostate cancer in African American patients, necessitating tailored screening and treatment approaches.
2. Health care access: Disparities in access to health care, particularly in underserved urban and in some rural areas, can influence the time to diagnosis and treatment. This can be exacerbated by a lack of health insurance, as well as well as a lower density of specialists and comprehensive cancer centers, and preclude access to standard approaches, clinical trials, and genetic testing.
3. Clinician bias: Implicit biases among clinicians can lead to disparities in the quality of care received by racial and ethnic minorities. This may affect treatment decisions, access to clinical trials as well as overall patient-clinician communication.

### How to Address Health Disparities in Prostate Cancer

Addressing these disparities requires a complex approach and certainly should be tailored for specific countries, states, or regions:

1. Improving access to care: Expanding health care access through policy reforms, community health initiatives, and increased funding for screening programs, particularly in underserved communities.
2. Culturally competent care: Training clinicians to deliver culturally sensitive care can help bridge gaps in communication and build trust with patients from different backgrounds.
3. Public awareness campaigns: Increasing awareness about prostate cancer risk factors, symptoms, and the importance of early detection can empower individuals to seek care and to engage in appropriate treatment options and trials.
4. Data collection: Continued research into the biological, environmental, and social determinants of health disparities in prostate cancer is essential to develop targeted interventions and policies that should be implemented in those underserved areas or populations.
5. Policy advocacy: Engaging in advocacy for equitable health care policies can help address systemic barriers that contribute to health disparities.

In 2023, ASCO updated its guideline protocol template to include an equity considerations section, allowing the panel to prioritize the needs of diverse populations prior to starting the literature search.

In conclusion, prostate cancer represents a global health challenge, even for high-income countries like the United States, characterized by profound health disparities and inequities that require attention. The understanding of the factors contributing to these disparities and the implementation of targeted strategies may allow health care systems to work toward equitable care for all patients at risk or affected by prostate cancer. The focus on these disparities not only improves individual outcomes but also enhances overall public health and equity in cancer care.

## MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient may have two or more such conditions—referred to as multiple chronic conditions (MCC)—is challenging. Patients with MCC are a complex and heterogeneous population, making it difficult to account for all of the possible permutations to develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude these patients in order to avoid potential interaction effects or confounding of results associated with MCC. As a result, the reliability of outcome data from these studies may be limited, thereby creating constraints for expert groups to make recommendations for care in this heterogeneous patient population.

As many patients for whom guideline recommendations apply present with MCC, any treatment plan needs to consider the complexity and uncertainty created by the presence of MCC and highlights the importance of shared decision making regarding guideline use and implementation. Therefore, in consideration of recommended care for the target index condition, clinicians should review all other chronic conditions present in the patient and take those conditions into account when formulating the treatment and follow-up plan.

In light of these considerations, practice guidelines should provide information on how to apply the recommendations for patients with MCC, perhaps as a qualifying statement for recommended care. This may mean that some or all of the recommended care options are modified or not applied, as determined by best practice in consideration of any MCC.

The typical patient with advanced prostate cancer is over 65 years and often has multiple chronic medical conditions including cardiovascular disease. By Medicare report, 66% of adults over 65 years have two or more chronic conditions and 16% have six or more.[50,51] The most common conditions in Medicare beneficiaries with prostate cancer are shown in Table 3 (multiple chronic conditions). Patients with prostate cancer who require systemic therapy with ADT, ARPIs, chemotherapy, and clinical trial therapies need to consider the possible impact of these therapies on drug-drug interactions, pill burden, and direct patient costs. These considerations may be overwhelming for a patient with newly diagnosed cancer, who is dealing with the stress of a cancer diagnosis, assembly of a care team, and care plan.

As noted previously, guidelines do not recommend the best treatment option for an individual patient, as an individual's optimal treatment will be guided by goals and preferences, their unique physiologic state, and life expectancy. Clinical tools have been developed to support decision making for older adults with multiple chronic conditions including those to identify patients' health priorities, to determine prognosis and health trajectory, and to consider deprescribing medications.[54]

ADT is universally prescribed in advanced prostate cancer and has cumulative side effects that can exacerbate common, concomitant medical conditions such as cardiovascular disease, osteoarthritis, obesity, diabetes, depression, and deconditioning. Additionally, ADT can reduce executive functioning and accelerate cognitive decline associated with aging thus complicating a patients' ability to manage multiple medical conditions. Optimizing a patient's prostate cancer care requires detailed medical history, patient education regarding the possible effects of ADT on chronic medical conditions, and specialist referral when appropriate to mitigate the possible negative effects of ADT. Caregiver and community support are also essential to optimizing prostate cancer outcomes and reducing treatment-related morbidity in the context of multiple medical conditions. See Table 3 for the most

**TABLE 3.** 10 Most Common Co-Occurring Chronic Conditions Among Male Medicare Beneficiaries With Prostate Cancer (N = 1,016,617), 2011

| Beneficiaries <65 Years (n = 26,243) | | Beneficiaries 65 Years and Older (n = 990,374) | |
|---|---|---|---|
| Condition | No. (%) | Condition | No. (%) |
| Prostate cancer prevalence, % | 0.9 | Prostate cancer prevalence, % | 8.8 |
| Top 10 co-morbidities | | Top 10 co-morbidities | |
| Hypertension | 17,767 (67.7) | Hypertension | 693,283 (70.0) |
| Hyperlipidemia | 13,589 (51.8) | Hyperlipidemia | 590,154 (59.6) |
| Diabetes | 9,982 (38.0) | Ischemic heart disease | 457,844 (46.2) |
| Ischemic heart disease | 9,291 (35.4) | Anemia | 337,957 (34.1) |
| Anemia | 8,478 (32.3) | Diabetes | 312,519 (31.6) |
| Arthritis | 7,833 (29.9) | Arthritis | 308,537 (31.2) |
| Chronic kidney disease | 6,286 (24.0) | Chronic kidney disease | 239,960 (24.2) |
| Depression | 6,162 (23.5) | Cataract | 235,241 (23.8) |
| COPD | 4,646 (17.7) | Heart failure | 203,231 (20.5) |
| Heart failure | 4,636 (17.7) | COPD | 145,818 (14.7) |

NOTE: Prepared by IPG/OIPDA on May 15, 2013. Data source: CMS administrative claims data, January 2011-December 2011, from the Chronic Condition Warehouse (CCW).[52] Population: Male Medicare beneficiaries enrolled in fee-for-service (FFS) coverage of both Parts A and B for the entire year. Beneficiaries who were enrolled at any point during the year in a Medicare Advantage (MA) plan were excluded as were beneficiaries who first became eligible for Medicare after January of the calendar year. Beneficiaries who died during the year were included up to their date of death if they meet the other inclusion criteria. Beneficiaries <65 years of age are primarily receiving Medicare due to a disability. Chronic condition measures: For these tables, chronic conditions were identified through Medicare administrative claims. Medicare beneficiaries were considered to have a chronic condition if the CMS administrative data had a claim indicating that they were receiving a service or treatment for the specific condition. The data tables include information for beneficiaries with one of 28 chronic conditions identified in the CCW. Detailed information on the identification of chronic conditions in the CCW is available at Chronic Conditions Data Warehouse: Home.[53] The list of 28 conditions include acquired hypothyroidism, acute myocardial infarction, Alzheimer's disease (including related disorders or senile dementia), anemia, asthma, atrial fibrillation, autism, benign prostatic hyperplasia, breast cancer, colon cancer, endometrial cancer, lung cancer, prostate cancer, cataract, chronic kidney disease, COPD, depression, diabetes, glaucoma, heart failure, hip/pelvic fracture, hypertension, hyperlipidemia, ischemic heart disease, osteoporosis, arthritis (OA and RA), schizophrenia (and other psychotic disorders), and stroke.
Abbreviations: CMS, Centers for Medicare & Medicaid Services; COPD, chronic obstructive pulmonary disease; IPG, Information Products Group; OIPDA, Office of Information Products and Data Analytics.

common comorbid conditions patients with prostate cancer present with.

## COST IMPLICATIONS

Increasingly, individuals with cancer are required to pay a larger proportion of their treatment costs through deductibles and coinsurance.[55,56] Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer treatments.[57,58]

Discussion of cost can be an important part of shared decision making.[59] Clinicians should discuss with patients the use of less expensive alternatives when it is practical and feasible for treatment of the patient's disease and there are two or more treatment options that are comparable in terms of benefits and harms.[59]

Table 4 (Costs) shows estimated prices for the available treatment options addressed in this guideline based on the average wholesale price. The costs listed in Table 4 are listed as per tablet/mL/etc. Thus, if prescribing enzalutamide at 160 mg once a day, and given a price of $143.32 US dollars (USD) per tablet, the daily price of four tablets a day would be $572 USD/day, or $17,160 USD for 30-day supply. Of note, medication prices may vary markedly in different markets, depending on negotiated discounts and rebates. Many of the costs of intravenous medications administered in a health care facility will be billed to insurance and covered differently than oral medications. Over the clinical course of mCRPC, many patients will undergo multiple lines of therapies compounding the lifetime costs of prostate cancer–directed therapy.

Patient out-of-pocket costs may further vary depending on insurance, including Medicare (with its $2,000 USD/year cap) coverage. Coverage may originate in the medical or pharmacy benefit, which may have different cost-sharing arrangements. Patients should be aware that different products may be preferred or covered by a particular insurance plan. Even for patients with the same insurance plan, the price may vary depending on the pharmacies they use. When discussing financial issues and concerns, patients should be made aware of any financial counseling services available to help navigate this complex and heterogeneous landscape.[59]

As part of the guideline development process, ASCO may opt to search the literature for published cost-effectiveness analyses that might inform the relative value of available treatment options. Excluded from consideration are cost-effective analyses that lack contemporary cost data; agents that are not currently available in either the United States or Canada; or agents that are industry-sponsored. No cost-effectiveness analyses were identified to inform the topic, likely in part given the continually changing landscape with many of the agents used earlier in the course of prostate cancer, in combination, or in sequence, making it difficult to guide therapy choices based on cost.

**TABLE 4.** Costs

| Drug | Dose | Form | AWP, USD | Price per |
|---|---|---|---|---|
| Abiraterone acetate[a] | 250 mg | Tablet | $1.60-97.21 | Each |
| Abiraterone acetate[a] | 500 mg | Tablet | $195.51-206.85 | Each |
| Abiraterone (micronized formulation) | 125 mg | Tablet | $104.57 | Each |
| Abiraterone acetate/niraparib | 50-500 mg, 100-500 mg | Tablet | $375.00 | Each |
| Apalutamide | 60 mg | Tablet | $149.13 | Each |
| Apalutamide | 240 mg | Tablet | $596.51 | Each |
| Darolutamide | 300 mg | Tablet | $135.74 | Each |
| Enzalutamide | 40 mg | Tablet, capsule | $143.32 | Each |
| Enzalutamide | 80 mg | Tablet | $286.65 | Each |
| Olaparib | 100 mg, 150 mg | Tablet | $168.54 | Each |
| Rucaparib | 200 mg, 250 mg, 300 mg | Tablet | $173.70 | Each |
| Talazoparib[b] | 0.1 mg, 0.25 mg, 0.35 mg, 0.5 mg | Capsule | $721.58 | Each |
| Sipuleucel- T | 50 million cells/250 mL | Suspension, intravenous | $300.49 | mL |
| $^{177}$Lu-PSMA (vipivotide tetraxetan) | 1,000 MBq/mL (27 mCi/1 mL) | Solution, intravenous | $54,621.00 | mL |
| Radium 223 ($^{223}$Ra-dichloride) | 1,100 kBq/mL (30 μCi/mL) | Solution, intravenous | $35,051.76 | mL |
| Docetaxel[a] | 10 mg/mL, 20 mg/mL | Solution, intravenous | $23.40-$365.15 | mL |
| Cabazitaxel | 60 mg/1.5 mL | Solution, intravenous | $11,317.23 | mL |
| Pembrolizumab | 100 mg/4 mL | Solution, intravenous | $1,700.61 | mL |

NOTE. Source—Lexicomp—UpToDate (Lexidrug, September 27, 2024).
Abbreviations: AWP, average wholesale price; USD, US dollars.
[a]Price range because of multiple manufacturers of generic.
[b]Would always have additional cost of enzalutamide.

## GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across various health care settings. Each ASCO guideline includes a community-based member on the Expert Panel (Dr Paul Celano and Dr Hamid Emamekhoo were the community practice representatives on this guideline panel). The additional role of this community-based clinician on the guideline panel is not only to assess the suitability of the recommendations for implementation in the community setting but also to identify any other barriers to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among front-line practitioners and survivors of cancer and caregivers, and also to provide adequate services in the face of limited resources. No barriers to implementation of the guideline recommendations were reported in the implementation survey for this guideline. The guideline recommendations table and accompanying tools (available at www.asco.org/genitourinary-cancer-guidelines) were designed to facilitate implementation of recommendations. This guideline will be distributed widely. ASCO guidelines are posted on the ASCO website and most often published in the *Journal of Clinical Oncology*.

### RELATED ASCO GUIDELINES

- Palliative Care for Patients With Cancer[60] (http://ascopubs.org/doi/10.1200/JCO.24.00542)
- Patient-Clinician Communication[38] (http://ascopubs.org/doi/10.1200/JCO.2017.75.2311)
- Bone Health and Bone-Targeted Therapies for Prostate Cancer[10] (https://ascopubs.org/doi/10.1200/JCO.19.03148)
- Systemic Therapy for Small-Cell Lung Cancer: ASCO-Ontario Health (Cancer Care Ontario) Guideline[11] (https://ascopubs.org/doi/10.1200/JCO.23.01435)
- Germline and Somatic Genomic Testing for Metastatic Prostate Cancer[8] (https://ascopubs.org/doi/10.1200/JCO-24-02608)

**ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.**

## ADDITIONAL RESOURCES

For current information, including selected updates, supplements, slide sets, and clinical tools and resources, visit www.asco.org/genitourinary-cancer-guidelines. The Data Supplement for this guideline includes additional supporting information. Guideline recommendations and algorithms are also available in the free ASCO Guidelines app (available for download in the Apple App Store and Google Play Store). Listen to key recommendations and insights from panel members on the ASCO Guidelines podcast. The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.org.

ASCO welcomes your comments on this guideline, including implementation challenges, new evidence, and how this guideline impacts you. To provide feedback, contact us at guidelines@asco.org. Comments may be incorporated into a future guideline update. To submit new evidence or suggest a topic for guideline development, complete the form available at www.asco.org/guidelines.

## GENDER-INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of individuals regardless of sexual orientation or gender identity.[61] Transgender and nonbinary people, in particular, may face multiple barriers to oncology care including stigmatization, invisibility, and exclusiveness. One way exclusiveness or lack of accessibility may be communicated is through gendered language that makes presumptive links between gender and anatomy.[62-65] With the acknowledgment that ASCO guidelines may impact the language used in clinical and research settings, ASCO is committed to creating gender-inclusive guidelines. The guideline authors strive to incorporate gender-inclusive language such as “people,” “individuals,” or “patients,” and omit reference to gender whenever possible. This includes statements referring to anatomy, as the authors acknowledge that transgender women have a prostate gland, for example. In instances in which the guideline draws upon data based on gendered research (eg, studies regarding men with prostate cancer), the authors describe the characteristics and results of the patient population and the research as reported.

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## Editor's Note

This ASCO Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, slide sets, clinical tools and resources, and links to patient information at www.cancer.org, is available at www.asco.org/genitourinary-cancer-guidelines.

## EQUAL CONTRIBUTION

R.G. and R.A.P. were Expert Panel Co-Chairs.

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

**Systemic Therapy in Patients With Metastatic Castration-Resistant Prostate Cancer: ASCO Guideline Update**

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**Stock and Other Ownership Interests:** Capio Biosciences, Archimmune Therapeutics, Nanorobotics
**Honoraria:** MJH Life Sciences, Aptitude Health, Curio Science, Peerview, Clinical Care Options, Mashup Media
**Consulting or Advisory Role:** Janssen, Exelixis, Pfizer, Bristol Myers Squibb, Seagen, Eisai, Aravive, Bayer, Lilly, AVEO, Merck, Sanofi/Aventis, Novartis, Gilead Sciences, EMD Serono, AstraZeneca, Loxo/Lilly, Xencor
**Research Funding:** Loxo/Lilly (Inst), Tempus (Inst), Pfizer (Inst), ALX Oncology (Inst), Exelixis (Inst), OncoC4, Astellas Pharma (Inst), Janssen (Inst), Janux Therapeutics (Inst), Bayer (Inst)
**Patents, Royalties, Other Intellectual Property:** Circulating tumor cell novel capture by c-MET technology (Inst), Prochelators as Targeted Prodrugs for Prostate Cancer (Inst)
**Travel, Accommodations, Expenses:** Bayer
**Other Relationship:** ASCO, Kidney Cancer Association, KidneyCan, Myrovlytis Trust, Winn Foundation

**Kathryn E. Beckermann**
**Consulting or Advisory Role:** Aravive, Exelixis, Bristol Myers Squibb, AVEO, Alpine Immune Sciences, Merck, AstraZeneca, Nimbus Therapeutics, Eisai, Arcus Biosciences, Xencor, Adicet Bio
**Research Funding:** Bristol Myers Squibb, ArsenalBio (Inst), Aravive (Inst), Pionyr (Inst)

**Michael A. Carducci**
**Honoraria:** Remedica
**Consulting or Advisory Role:** Pfizer, Acrivon Therapeutics, AstraZeneca, Sanofi
**Research Funding:** Pfizer (Inst), Arcus Biosciences (Inst), Merck (Inst), Celgene/Bristol Myers Squibb (Inst)

**Sebastien J. Hotte**
**Honoraria:** Astellas Scientific and Medical Affairs Inc, Bayer, Advanced Accelerator Applications/Novartis
**Consulting or Advisory Role:** Janssen, Astellas Pharma, Pfizer, Bayer, Merck, Novartis Canada Pharmaceuticals Inc
**Research Funding:** AstraZeneca (Inst), Merck (Inst), Bristol Myers Squibb (Inst), Janssen Oncology (Inst), Ayala Pharmaceuticals (Inst), Pfizer (Inst), Astellas Pharma (Inst), Bristol Myers Squibb (Inst), Seagen (Inst), SignalChem (Inst), ESSA (Inst)

**Arnab Basu**
**Honoraria:** Gilead Sciences, Cardinal Health, Eisai, Natera
**Consulting or Advisory Role:** EMD Serono, Seagen, Bristol Myers Squibb/Pfizer
**Speakers' Bureau:** Eisai
**Research Funding:** Merck (Inst), EMD Serono (Inst), Natera (Inst), Astellas Pharma (Inst), Bristol Myers Squibb/Celgene (Inst), Genentech/Roche (Inst), AVEO (Inst)

**Hala Borno**
**Employment:** Collective Health, Waymark, Trial Library
**Leadership:** Collective Health, Waymark, Trial Library
**Stock and Other Ownership Interests:** Collective Health, Waymark, Trial Library
**Honoraria:** Dendreon, BMS, ConcertAI, AstraZeneca
**Consulting or Advisory Role:** Dendreon, BMS, AstraZeneca

**Alan H. Bryce**
**Honoraria:** Pfizer, Janssen, Novartis, Verity Pharmaceuticals, Astellas Pharma, Bayer, Pfizer, Janssen Oncology, Lilly, Lantheus Medical Imaging, MOMA Therapeutics, Tolmar, Pfizer
**Consulting or Advisory Role:** Astellas Pharma
**Travel, Accommodations, Expenses:** Pfizer (Inst), Bayer, Pfizer, Tolmar

**Elizabeth Wulff-Burchfield**
**Stock and Other Ownership Interests:** Nektar, Immunomedics
**Honoraria:** OncologyLive
**Consulting or Advisory Role:** Astellas Scientific and Medical Affairs Inc, Exelixis, Bristol Myers Squibb, Janssen Oncology, AVEO, Aptitude Health, Targeted Oncology
**Research Funding:** Pfizer, Acerta Pharma/AstraZeneca (Inst)
**Patents, Royalties, Other Intellectual Property:** Methods of use patent application has been submitted for the pharmaceutical product Osanetant, owned by Acer Therapeutics
**Travel, Accommodations, Expenses:** Exelixis

**Lisa Bodei**
**Consulting or Advisory Role:** Wren Laboratories
**Research Funding:** Advanced Accelerator Applications/Novartis (Inst), Molecular Targeting Technologies (Inst), AAA/Endocyte/Novartis (Inst), AAA/Endocyte/Novartis (Inst), ITM Isotope Technologies Munich (Inst), POINT Biopharma (Inst), RayzeBio (Inst), Abdera Thereapeutics (Inst), Fusion Pharmaceuticals (Inst)
**Uncompensated Relationships:** AAA/Endocyte/Novartis, ITM Isotope Technologies Munich, IBA RadioPharma Solutions, Clovis Oncology, Molecular Targeting Technologies, RayzeBio, POINT Biopharma, Fusion Pharmaceuticals, Amgen, Great Point Partners

**Andrew Loblaw**
**Employment:** Sanofi
**Stock and Other Ownership Interests:** Sanofi
**Honoraria:** Astellas Pharma, Bayer, Tersera, Knight Therapeutics, Tolmar, Janssen Oncology, Sumitomo Pharma Oncology
**Consulting or Advisory Role:** Janssen Oncology, Bayer, TerSera, Astellas Pharma
**Research Funding:** Tersera
**Patents, Royalties, Other Intellectual Property:** Prostate immobilization device (GU-Lok)
**Travel, Accommodations, Expenses:** Tolmar

**Robert J. Hamilton**
**Honoraria:** Janssen, Bayer, Astellas Pharma, Merck, Knight Pharmaceuticals, Tolmar, Tersera, Novartis
**Research Funding:** Janssen, Bayer, Novartis
**Travel, Accommodations, Expenses:** Janssen, Tolmar

**Hamid Emamekhoo**
**Consulting or Advisory Role:** Janssen Biotech, AVEO, Cardinal Health, Eisai, Bristol Myers Squibb company, Janssen Biotech, Cardinal Health

**Thomas A. Hope**
**Stock and Other Ownership Interests:** RayzeBio, Curium Pharma
**Honoraria:** GE Healthcare, Cardinal Health
**Consulting or Advisory Role:** Ipsen, Curium Pharma, Blue Earth Diagnostics, Bayer, ITM Isotope Technologies Munich, Lantheus Medical Imaging, Sanofi
**Research Funding:** GE Healthcare (Inst), Philips Healthcare (Inst), Advanced Accelerator Applications (Inst), Clovis Oncology (Inst), Lantheus Medical Imaging (Inst), Janssen Oncology (Inst), Telix Pharmaceuticals (Inst), Novartis (Inst), Bayer (Inst)
**Travel, Accommodations, Expenses:** GE Healthcare

**M. Hassan Murad**
**Research Funding:** American Urological Association

**Hongfang Liu**
**Employment:** Mayo Clinic

**Rahul A. Parikh**
**Stock and Other Ownership Interests:** IBRX
**Patents, Royalties, Other Intellectual Property:** Patents on DNA repair pathways in cancer. I do not receive any royalties for the same

No other potential conflicts of interest were reported.

## APPENDIX 1. GUIDELINE DISCLAIMER

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## APPENDIX 2. GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at http://www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

**TABLE A1.** Systemic Therapy in Patients With Metastatic Castration-Resistant Prostate Cancer Guideline Expert Panel Membership

| Name | Affiliation | Area of Expertise |
|---|---|---|
| Rohan Garje, MD, Co-Chair | Miami Cancer Institute, Baptist Health South Florida, Miami, FL | Medical oncology |
| Rahul Parikh, MBBS, PhD, Co-Chair | University of Kansas Medical Center, Westwood, KS | Medical oncology |
| Arnab Basu, MBBS, MPH | University of Alabama at Birmingham, Birmingham, AL | Medical oncology |
| Kathryn E. Beckermann, MD, PhD | Vanderbilt University School of Medicine, Nashville, TN | Medical oncology |
| Lisa Bodei, PhD, MD | Weill Cornell Medical College of Cornell University and Memorial Sloan Kettering Cancer Center, New York, NY | Nuclear medicine |
| Hala Borno, MD | University of California, San Francisco, San Francisco, CA | Medical oncology |
| Alan H. Bryce, MD | City of Hope, Goodyear, AZ | Medical oncology |
| Michael A. Carducci, MD, FASCO | Sidney Kimmel Cancer Center, Johns Hopkins University, Baltimore, MD | Medical oncology |
| Paul Celano, MD, FASCO | The Cancer Center at GBMC, Towson, MD | Medical oncology |
| Hamid Emamekhoo, MD | University of Wisconsin, Madison, WI | Medical oncology |
| Robert J. Hamilton, MD, MPH (OH(CCO) panel rep) | Cancer Clinical Research Unit, Princess Margaret Cancer Centre, Toronto, ON, Canada | Urology |
| Huan He, PhD | Yale University, New Haven, CT | Epidemiology/informatics |
| Daniel Herchenhorn, MD, PhD | Instituto D'Or/Oncologia D'Or, Latin America Cooperative Group (LACOG) - Genito-Urinary, Rio de Janeiro, Brazil | Medical oncology |
| Thomas A. Hope, MD | University of California, San Francisco, San Francisco, CA | Interventional/diagnostic radiology |
| Sebastien J. Hotte, MD, MSc (OH(CCO) panel rep) | McMaster University, Hamilton, ON, Canada | Medical oncology |
| Terry M. Kungel, MBA | Woolwich, ME | Patient representative |
| Hongfang Liu, PhD | Mayo Clinic Hospital, Phoenix, AZ | Epidemiology/informatics |
| Andrew Loblaw, MD, FASCO (OH(CCO) panel rep) | Odette Cancer Centre, Sunnybrook Health Sciences Centre, Toronto, ON, Canada | Radiation oncology |
| M. Hassan Murad, MD, MPH | Mayo Clinic, Rochester, MN | Epidemiology/informatics |
| Syed Arsalan Ahmed Naqvi, MD | Mayo Clinic, Phoenix, AZ | Medical oncology |
| Irbaz Bin Riaz, MD, PhD | Mayo Clinic, Phoenix, AZ | Medical oncology/informatics |
| Mary-Ellen Taplin, MD, FASCO | Dana-Farber Cancer Institute, Boston, MA | Medical oncology |
| Neha Vapiwala, MD | University of Pennsylvania Abramson Cancer Center, Philadelphia, PA | Radiation oncology |
| Peng Wang, MD, PhD | Ohio State University Wexner Medical Center, Columbus, OH | Medical oncology |
| James Elbert Williams Jr., MS | SPHR Pennsylvania Prostate Cancer Coalition, Camp Hill, PA | Patient representative |
| Elizabeth Wulff-Burchfield, MD | University of Kansas Medical Center, Kansas City, KS | Medical oncology |
| Tian Zhang, MD, MHS | Simmons Comprehensive Cancer Center, UT Southwestern Medical Center, Dallas, TX | Medical oncology |
| R. Bryan Rumble, MSc | American Society of Clinical Oncology, Alexandria, VA | ASCO practice guideline staff (health research methods) |

**TABLE A2.** Recommendation Rating Definitions

| Term | Definition |
|---|---|
| Quality of evidence | |
| High | We are very confident that the true effect lies close to that of the estimate of the effect |
| Moderate | We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different |
| Low | Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect |
| Very low | We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect |
| Strength of recommendation | |
| Strong | In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects |
| | In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects |
| | All or almost all informed people would make the recommended choice for or against an intervention |
| Conditional/weak | In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists |
| | In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists |
| | Most informed people would choose the recommended course of action, but a substantial number would not |

NOTE. GRADE Handbook, Schünemann et al.[7]